

**DEVELOPMENT OF BRIDGE TOOLS FOR LIGAND DATABASE**  
**DOCKING: DEMONSTRATED ON DENGUE TARGETS**

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University of Sri Jayewardenepura, Sri Lanka.

BSc. Hons (Chem) Degree

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# **DEVELOPMENT OF BRIDGE TOOLS FOR LIGAND DATABASE**

## **DOCKING: DEMONSTRATED ON DENGUE TARGETS**

A dissertation submitted

to

The Department of Chemistry of the

University of Sri Jayewardenepura

in partial fulfillment of the

requirement for the

Bachelor of Science Honors Degree

in Chemistry

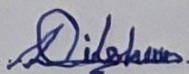
by

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## DECLARATION

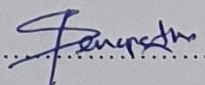
The work described in this dissertation was carried out by me in the Department of Chemistry, University of Sri Jayewardenepura, Sri Lanka, under the guidance of Dr. Tharindu Senapathi has not been submitted elsewhere.

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## **DEDICATION**

To Amma and Thaththa

## **ACKNOWLEDGEMENT**

First and foremost, I would like to express my sincere gratitude to my research supervisor, Dr. Tharindu Senapathi, Senior Lecturer in the Department of Chemistry, University of Sri Jayewardenepura, for his continuous guidance and encouragement throughout the completion of my research project. Without his valuable comments, suggestions, and generous help, this project would not have been concluded successfully. I would also like to express my deep appreciation and gratitude to Prof. N. T. Perera, Head of the Department of Chemistry, University of Sri Jayewardenepura, for their endless cooperation throughout the project.

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## ABBREVIATIONS

|           |                                                                   |
|-----------|-------------------------------------------------------------------|
| Å         | Ångström                                                          |
| ABFE      | Absolute Binding Free Energy                                      |
| AcetylCoA | Acetyl coenzyme A                                                 |
| ALA       | Alanine                                                           |
| ASN       | Asparagine                                                        |
| BRIDGE    | Biomolecular Reaction and Interaction Dynamics Global Environment |
| CADD      | Computer-Aided Drug Design                                        |
| CHARMM    | Chemistry at HARvard Macromolecular Mechanics                     |
| CPU       | Central Processing Unit                                           |
| CSV       | Comma-separated values                                            |
| DENV      | Dengue virus                                                      |
| EMA       | European Medicines Agency                                         |
| FDA       | Food and drug administration                                      |
| GLN       | Glutamine                                                         |
| GUI       | Graphical User Interface                                          |
| HTTP      | Hypertext Transfer Protocol                                       |
| LBDD      | Ligand-Based Drug Design                                          |
| LYS       | Lysine                                                            |
| MD        | Molecular Dynamics                                                |
| MM/GBSA   | Molecular Mechanics/Generalized Born Surface Area                 |
| MM/PBSA   | Molecular Mechanics-Poisson-Boltzmann Surface Area                |

|      |                                              |
|------|----------------------------------------------|
| PDB  | Protein Data Bank                            |
| PDB  | Protein Data Bank                            |
| PLIP | Protein-Ligand Interaction Profiler          |
| QSAR | Quantitative structure-activity relationship |
| RMSD | Root Mean Square Deviation                   |
| RNA  | Ribonucleic acid                             |
| SASA | Solvent-accessible surface area              |
| SBDD | Structure-Based Drug Design                  |
| SER  | Serine                                       |
| THR  | Threonine                                    |
| TXT  | Text file                                    |
| VAL  | Valine                                       |
| XML  | eXtensible Markup Language                   |



## TABLE OF CONTENTS

|                                                                 |      |
|-----------------------------------------------------------------|------|
| DECLARATION .....                                               | iii  |
| DEDICATION.....                                                 | iv   |
| ACKNOWLEDGEMENT .....                                           | v    |
| ABBREVIATIONS .....                                             | vi   |
| TABLE OF CONTENTS .....                                         | viii |
| LIST OF TABLES.....                                             | x    |
| LIST OF FIGURES.....                                            | xi   |
| ABSTRACT .....                                                  | xiv  |
| INTRODUCTION.....                                               | 1    |
| 1.1 Drug Discovery.....                                         | 1    |
| 1.2 Computer-Aided Drug Discovery. ....                         | 2    |
| 1.3 Molecular docking in drug discovery .....                   | 4    |
| Section A: Development of Galaxy BRIDGE Tool.....               | 7    |
| 2.1 Galaxy BRIDGE Tool .....                                    | 8    |
| 2.2 Python Script(docking.py).....                              | 16   |
| 2.3 XML Wrapper File .....                                      | 24   |
| 2.4 Benchmarking Galaxy BRIDGE Tool: RMSD-Based Evaluation..... | 30   |
| 2.5 Result and Discussion.....                                  | 32   |
| 2.6 Conclusion .....                                            | 38   |
| Section B: Application of the Galaxy BRIDGE Tool.....           | 39   |
| 3.1 Dengue Virus .....                                          | 40   |
| 3.2 Natural products .....                                      | 42   |
| 3.3 Natural product database .....                              | 44   |

|                                                                                                                                                        |     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 3.4 Computational Methods .....                                                                                                                        | 50  |
| 3.5 Molecular Dynamics Simulation. ....                                                                                                                | 51  |
| 3.6 Explanation Of Binding Free Energy Calculation Using Molecular Mechanics and<br>Solvation Models, Including Polar and Nonpolar Contributions. .... | 53  |
| 3.7 Results and Discussion .....                                                                                                                       | 57  |
| 3.8 Analysis of binding free energy calculations (MM/GBSA, MM/PBSA).....                                                                               | 60  |
| 3.9 RMSD Calculation .....                                                                                                                             | 62  |
| 3.10 H-Bond Analysis .....                                                                                                                             | 87  |
| 3.11 Toxicity Predictions.....                                                                                                                         | 90  |
| 3.12 Conclusion .....                                                                                                                                  | 95  |
| REFERENCES.....                                                                                                                                        | 96  |
| APPENDIX .....                                                                                                                                         | 100 |
| DATA AVAILABILITY .....                                                                                                                                | 105 |

## LIST OF TABLES

|                                                                                                                                                                                                                     |    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Table 2.1.</b> Overview of the scoring functions and search algorithms .....                                                                                                                                     | 9  |
| <b>Table 2.2.</b> Comparison of standard and docked binding residues for various protein-ligand complexes, along with their corresponding Root Mean Square Deviation (RMSD) values indicating docking accuracy..... | 32 |
| <b>Table 3.1</b> Docking scores and binding free energy calculations (MMGBSA and MMPBSA) for the ligand candidates targeting the dengue virus NS2B-NS3 protease. ....                                               | 59 |
| <b>Table 3.3</b> Table of H-bond analysis of the Molecular Dynamics simulation of protein and ligand complexes. ....                                                                                                | 88 |
| <b>Table 3.4</b> Description of toxicity classes.....                                                                                                                                                               | 91 |
| <b>Table 3.5</b> List of molecules with their Molecular IDs, chemical structures, and corresponding predicted toxicity classes. ....                                                                                | 92 |

## LIST OF FIGURES

|                                                                                                                                                                                                                                                                                                                                                   |    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Figure 1.1</b> A schematic representation of virtual screening approaches in drug discovery showing the hits identification process in CADD.....                                                                                                                                                                                               | 3  |
| <b>Figure 1.2</b> A general workflow of the molecular docking-based drug discovery process. ....                                                                                                                                                                                                                                                  | 5  |
| <b>Figure 2.1</b> A histogram of the RMSD values fitted to a skewed distribution. The x- and y-axes correspond to RMSD values and their frequency, respectively. The distribution of the RMSD values is shown as the blue curve. The red dashed vertical line represents a particular RMSD threshold.....                                         | 34 |
| <b>Figure 2.1.</b> Structural alignment of ligand poses within binding sites from two protein-ligand complexes. <b>(A)</b> shows that the complex 1abeA_BS01_UUU has an RMSD of 0.044. <b>(B)</b> shows the complex 1kwfA_BS01_BGC with an RMSD of 6.354. Protein backbones are shown in cyan, ligands in red (reference) and blue (target). .... | 35 |
| <b>Figure 3.1</b> Illustration of DENV structural and NS proteins on the ER with their cleavage sites .....                                                                                                                                                                                                                                       | 41 |
| <b>Figure 3.2</b> Gaussian Calculation Setup used to create Natural Product Database. ....                                                                                                                                                                                                                                                        | 46 |
| <b>Figure 3.3</b> RMSD analysis of the 10 ns molecular dynamics simulation with salianic acid <b>(A)</b> . The time series plot of RMSD of the protein backbone (black) and ligand (red). Histograms display the distribution of RMSD values for the backbone <b>(B)</b> and ligand <b>(C)</b> . ....                                             | 64 |
| <b>Figure 3.4</b> The initial conformation of salianic acid at 1 <sup>st</sup> nanosecond (red) and its repositioned conformation at 10 <sup>th</sup> nanoseconds (dark blue) within the protein binding pocket.....                                                                                                                              | 65 |
| <b>Figure 3.5</b> RMSD analysis of the 10 ns molecular dynamics simulation with Benzyliothiocyanate <b>(A)</b> . The time series plot of RMSD of the protein backbone (black) and                                                                                                                                                                 |    |

ligand (red). Histograms display the distribution of RMSD values for the backbone **(B)** and ligand **(C)**. ..... 67

**Figure 3.6** The initial conformation of Benzylisothiocyanate at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket. .... 68

**Figure 3.7** RMSD analysis of the 10 ns molecular dynamics simulation with Erysolin **(A)**. The time series plot of RMSD of the protein backbone (black) and ligand (red). Histograms display the distribution of RMSD values for the backbone **(B)** and ligand **(C)**. ..... 70

**Figure 3.8** The initial conformation of Erysolin at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket. .... 71

**Figure 3.9** RMSD analysis of the 10 ns molecular dynamics simulation with cinnamaldehyde **(A)**. The time series plot of RMSD of the protein backbone (black) and ligand (red). Histograms display the distribution of RMSD values for the backbone **(B)** and ligand **(C)**. .. 73

**Figure 3.10** The initial conformation of Cinnamaldehyde at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket. .... 74

**Figure 3.11** RMSD analysis of the 10 ns molecular dynamics simulation with cinnamyl alcohol **(A)**. The time series plot of RMSD of the protein backbone (black) and ligand (red). Histograms display the distribution of RMSD values for the backbone **(B)** and ligand **(C)**. .. 76

**Figure 3.12** The initial conformation of Cinnamyl alcohol at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket. .... 77

**Figure 3.13** RMSD analysis of the 10 ns molecular dynamics simulation with castanospermine (A). The time series plot of RMSD of the protein backbone (black) and ligand (red).

Histograms display the distribution of RMSD values for the backbone (B) and ligand (C).. 79

**Figure 3.14** The initial conformation of Castanospermine at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket.

..... 80

**Figure 3.15** RMSD analysis of the 10 ns molecular dynamics simulation with cinnamic acid(A). The time series plot of RMSD of the protein backbone (black) and ligand (red).

Histograms display the distribution of RMSD values for the backbone (B) and ligand (C).. 82

**Figure 3.16** The initial conformation of Cinnamic Acid at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket.

..... 83

**Figure 3.17** RMSD analysis of the 10 ns molecular dynamics simulation with sulforaphane (A). The time series plot of RMSD of the protein backbone (black) and ligand (red).

Histograms display the distribution of RMSD values for the backbone (B) and ligand (C).. 85

**Figure 3.18** The initial conformation of Sulforaphane at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket.

..... 86

## ABSTRACT

Dengue fever is a critical public health problem with no specific drug currently available. This study aims to identify potential inhibitors for the NS2B-NS3 protease (PDB ID: 2FOM) of the dengue virus using a Computer-Aided Drug Discovery method. The Galaxy BRIDGE tool was developed based on the AutoDock Vina docking engine to find a drug for the dengue virus and integrated into the Galaxy web-based platform. This tool provides accessibility, reproducibility, shareability, and transparency with a user-friendly interface. To confirm the docking protocol, benchmark testing was done by using a dataset of experimentally validated compounds. The results confirmed that AutoDock Vina gives reliable and accurate predictions for most protein-ligand complexes, supporting its use in virtual screening pipelines.

In parallel, a natural product database was developed, consisting of dengue virus-related ligands sourced from PubChem and optimized using the semi-empirical PM6 method. These compounds were docked against the dengue virus NS2B-NS3 protease using the developed Galaxy BRIDGE tool. Then the top-ranked complexes were subjected to molecular dynamics (MD) simulations followed by MM/GBSA and MM/PBSA free energy calculations. RMSD analysis was performed to assess structural stability, and hydrogen bond analysis, and *in silico* toxicity evaluations were done to assess drug-likeness and safety. The results show significant differences in binding energy after rescoring, emphasizing the importance of including solvation effects and conformational flexibility in accurate binding affinity estimation. This integrated pipeline shows the value of combining natural product databases with reproducible, open-source computational tools for drug discovery against the dengue virus.

# CHAPTER 1

## INTRODUCTION

### 1.1 Drug Discovery

Drug discovery is a vital and complex process that leads to the development of new medicines for the treatment, cure, and prevention of diseases. It involves clearly defined phases based on biology, chemistry, pharmacology, and medicine. The approach starts with a molecular characterization of the disease. Researchers first identify the biological processes that drive the disease to find a specific target, such as a protein, enzyme, or receptor. It is likely to play a central role in how the disease progresses. That's target identification, which is then followed by target validation. Scientists know that altering the target can work therapeutically.

After a viable target has been clearly identified, the next challenge is to identify small-molecule chemical structures that can bind the target and modulate its function. These hits can be discovered using high-throughput screening of vast chemical libraries or by exploring natural products. The active hits are then identified as lead compounds, and lead optimization is conducted with related chemical modifications. Here, the emphasis of this phase is to increase activity, decrease side effects, and increase the properties of the medicines to be absorbed and circulated in the body. At this point, it is possible to synthesize several analogues of the lead compound and test their activity to determine which works best.

Ahead of the optimization procedures, the final compound undergoes several preclinical tests, during which laboratory experiments and animal studies are conducted to estimate its toxicity, safety profile, optimal dosage, pharmacokinetics (i.e., how a drug moves



through an organism), and pharmacodynamics (i.e., how a drug affects an organism). With perseverance, a compound enters clinical testing, which consists of three core phases. First, in Phase 1, a small group of healthy individuals is engaged to test the safety and dosage of the drug. In Phase 2, a small group of patients estimates how a drug affects the human body. Phase three involves a larger group of human participants and compares a new treatment to existing ones. All stages are strictly regulated and require high standards of safety and exemplarity.<sup>1</sup>

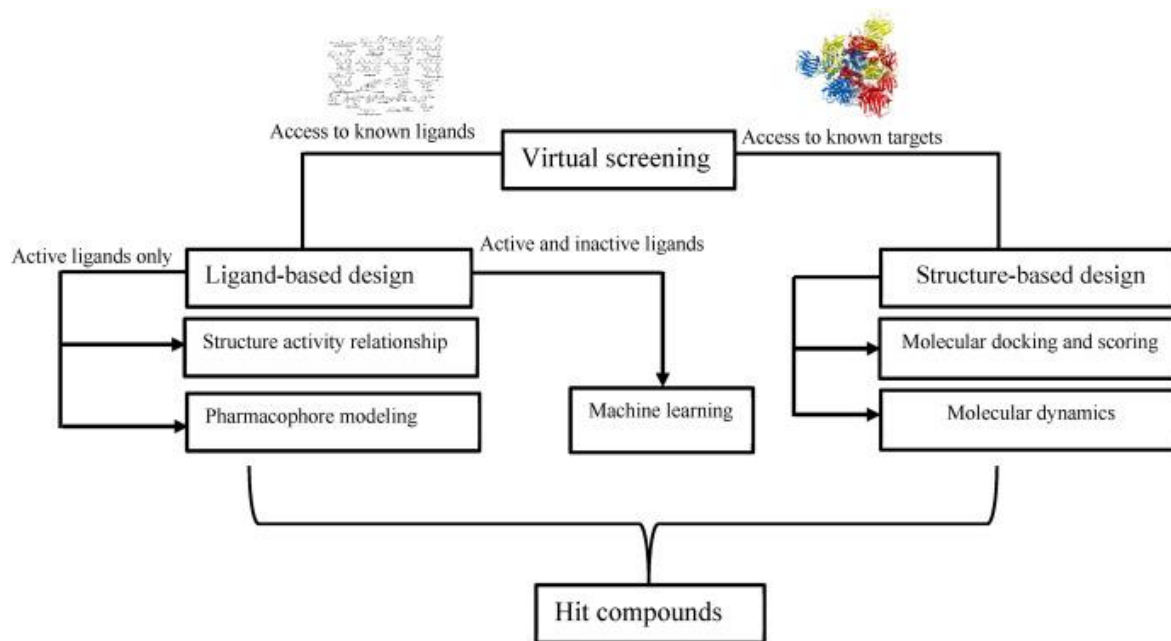
After study results have been successfully demonstrated in the clinical phase, the drug developer reports the findings to the regulatory authorities (e.g., FDA or EMA). Upon approval, the drug is introduced into the market at the post-marketing surveillance (Phase IV) level to spot any long-term and rare side effects that were not detected during clinical trials.

The discovery of new drugs is extremely expensive and time-consuming, typically takes 10-15 years to complete and involves a significant amount of collaboration in science and finance. Drug discovery is a daunting challenge due to its complexity, but it is necessary for advancing medicine, as new and improved medical treatments can have a significant impact on the quality of life for people around the world.<sup>2, 3</sup>

## **1.2 Computer-Aided Drug Discovery.**

Drug discovery is a challenging and expensive process that relies on unexpected discoveries and results from experimental screening. Nevertheless, the development of high computational power and molecular modeling applications brought Computer-Aided Drug Discovery (CADD) to the forefront of modern pharmaceutical research. Through the application of in silico screening, CADD generated potential drug candidates to facilitate the

excitation from traditional drug research-long time and non-negligible monies can be greatly reduced.



**Figure 1.1** A schematic representation of virtual screening approaches in drug discovery showing the hits identification process in CADD.<sup>4</sup>

Computer-Aided Drug Design (CADD) encompasses a series of computational methods that can be categorized into two main types based on the methods used: Structure-Based Drug Design (SBDD) and Ligand-Based Drug Design (LBDD) (see Figure 1.1). In SBDD, the knowledge of the three-dimensional structure of a biological target (usually a protein) is exploited to develop or find compounds that bind to the target. In comparison, LBDD is based on the activity of known compounds to impart activity to new molecules where the target structure is unknown.<sup>4-6</sup>

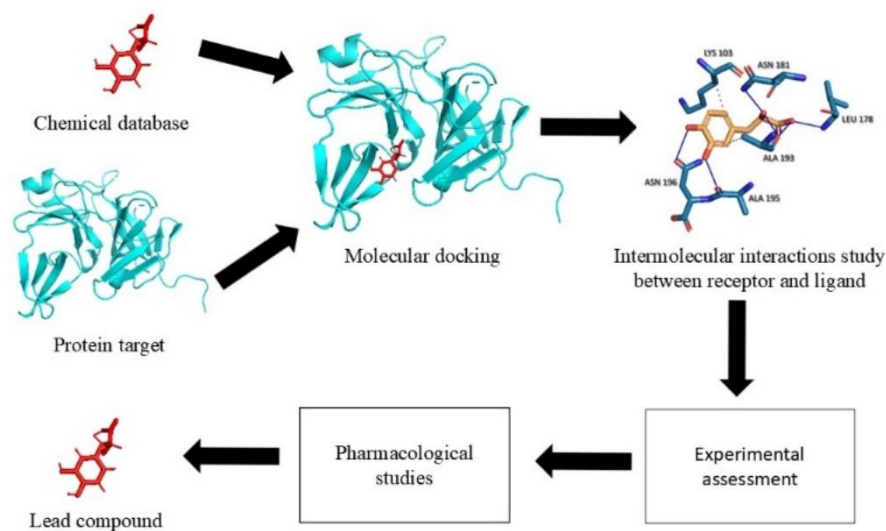
The major steps of the CADD process involve target identification and validation, molecular docking, virtual screening, pharmacophore modeling, quantitative structure-activity

relationship (QSAR) analysis, and molecular dynamics (MD) simulation. Such methods aid in the prediction of the binding affinity, specificity, or stability of drug candidates and thus enable the selection of molecules with a maximum therapeutic benefit.<sup>6, 7</sup>

Incorporation of CADD early in the drug development process enables faster lead identification, increases hit-to-lead optimization, and lessens the chances of failure in the later stages of development. The applicability of which is particularly important in natural product drug discovery, antiviral study, and precision medicine.

### **1.3 Molecular docking in drug discovery**

Molecular docking is an essential approach in structure-based drug discovery, which can predict the binding patterns of small molecules (ligands) into the active site of a biomacromolecule, such as an enzyme or receptor.<sup>8</sup> By predicting the optimal orientation of the ligand within the binding site of the target protein, along with the relative contribution of its protein-ligand interaction energy, docking assists in the identification of molecules worthy of experimental validation. This strategy is often adopted in virtual screening, lead optimization, and target validation, leading to a time-saving and significant improvement in the efficiency of early-stage drug discovery. Despite its increasing popularity, the broader adoption of molecular docking is limited by major accessibility, reproducibility, shareability, and transparency issues.<sup>9-11</sup>



**Figure 1.2** A general workflow of the molecular docking-based drug discovery process.

First, most docking tools are either commercial programs or demand considerable computational expertise and resources, and they are generally not available to researchers working in resource-poor environments. Even in the case of open-source alternatives, the learning curve can be steep due to difficult installations or interfaces that are not user-friendly. Another major issue is reproducibility, as docking results can vary depending on the software used, its version, parameters, scoring function, and ligand/protein preparation. Frequently, the necessary conditions for exactly reproducing the results are not reported in such studies. Furthermore, the barrier to sharing docking workflows and data remains substantial; although publications may present summary tables and images of predicted poses, they rarely provide the input files, setups, and output results required to verify and reuse results. This inhibits research collaboration and the building of openly reusable scientific knowledge. Lastly, the failure to report computational methods transparently, such as inadequately detailed or missing procedural information, also erodes confidence in docking-derived results.<sup>9</sup>

Overcoming these challenges is crucial for molecular docking to achieve its full potential in terms of computational performance and significantly accelerate drug design. Making docking workflows more accessible, reproducible, shareable, and transparent would contribute to more reliable and collaborative research, bringing the field closer to the goals of open science and data-driven innovation.

In this study, a database of natural products from Sri Lankan plants was created and used to search for potential drug-like compounds against the dengue virus. Section A focuses on describing the Galaxy BRIDGE tool, while Section B addresses the application of the tool developed in Section A. Finally, molecular dynamics simulations are applied to the top 10 ligands obtained from the docked 2FOM protein and natural product database to identify promising drug compounds against the dengue virus.

**Section A:**

**Development of Galaxy BRIDGE  
Tool**

## CHAPTER 2

### 3.1 Galaxy BRIDGE Tool

The Galaxy BRIDGE tool was developed to integrate and streamline various computational drug discovery processes within the Galaxy platform. It serves as a pipeline for integrating various software packages, such as molecular docking and data analysis tools, into a single, user-friendly set of applications. The development was designed to automate complex procedures, increase the ability to repeat results, and provide access to cutting-edge drug discovery simulations for non-computer experts. By utilizing XML files as wrappers and Python scripts for execution, Galaxy BRIDGE streamlines the management of docking jobs and data handling, thereby making the drug discovery process more accessible and efficient.

The tool is automated and designed to facilitate the molecular docking process. It can be used to model and study the binding interactions of small molecules and ligands with the target protein. The highly publicized AutoDock Vina, a molecular docking engine known for fast and reliable docking simulations, has been used to develop this Galaxy BRIDGE tool.

AutoDock Vina<sup>13, 14</sup> is a popular molecular docking program that predicts the binding mode of small molecules, such as a ligand bound to a macromolecule, typically a protein. It is a valuable method for in-silico drug design, as it allows for the virtual screening of large compound databases and provides binding poses and estimated binding affinities. AutoDock Vina improves upon its predecessor, AutoDock 4, and although both are used as frequently as

each other in the published literature, Vina has been shown to significantly outperform AutoDock 4 in terms of accuracy and efficiency.<sup>15</sup>

**Table 2.1.** Overview of the scoring functions and search algorithms.<sup>16</sup>

| <b>Program</b> | <b>Scoring Function</b>    | <b>Search Algorithm</b>       |
|----------------|----------------------------|-------------------------------|
| AutoDock Vina  | Empirical+ Knowledge-based | Random perturbation           |
| PLANTS         | Empirical                  | Ant colony optimization       |
| Gold           | Empirical + Physics-based  | Genetic Algorithm             |
| Glide          | Empirical + Physics-based  | Hierarchical search           |
| Molegro        | Empirical + Physics-based  | Guided differential evolution |
| Autodock4(AD4) | Empirical + Physics-based  | Lamarckian Genetic algorithm  |
| rDock          | Empirical                  | GA, simulated annealing       |

Table 2.1 shows that the scoring functions are generally placed into three main categories: physics-based SFs, which typically take the force field approach from molecular mechanics; empirical SFs, which use statistical regression to fit (sometimes arbitrary) functions to some experimental data (eg Ki or Kd or IC50 value); and knowledge-based SFs, which use all other data, e.g. covalent bond lengths and statistical distribution of the interatomic distances determined from many crystal structures. In reality, many scoring functions incorporate a mix of these to some extent. For instance, AutoDock 4 (AD4)<sup>17</sup> predominantly applies a physics-based scoring function (SF) with several terms from the Amber family of force fields, but scaled to experimental binding constants based on their hydrophobic and non-hydrophobic components, thus constituting a possible description as a hybrid physics-based/empirical scoring function. In contrast, AutoDock Vina (hereafter ‘Vina’) relies on an empirical SF based



on a five-term Piece-wise linear potential (PLP); however, the incorporation of atomic van der Waals radii in the SF means that Vina might be regarded as a hybrid empirical/knowledge-based SF. The use of PLPs in empirical scoring functions is widespread because fitting constant scaling factors for the terms could be achieved easily with multiple linear regression.<sup>16</sup>

At the center of AutoDock Vina is a scoring function to estimate the binding affinity of the ligand with the target protein. The score function estimates the binding free energy,  $\Delta G_{\text{bind}}$ , which reflects the strength of the ligand-protein interaction as well as its stability. Vina employs an empirical scoring function that describes a weighted sum of various types of physical interactions.

The empirical scoring function must be performed to cumulate individual contributions towards the affinity, or fitness, of protein-ligand binding. The terms generally describe a significant energetic contribution to the binding of a protein and a ligand. Each of these functions has several parameters that can be trained to improve the predictions. Finally, those terms are all weighted and summed (multiplied by a constant) to get the predicted binding affinity. The binding energy is predicted as the sum of distance-dependent atom pair interactions.

$$E = \sum e_{\text{pair}}(d) \quad (2.1)$$

$$d = r - R_i - R_j \quad (2.2)$$

In Equation 2.1 the surface distance is calculated, where  $r$  is the interatomic distance and  $R_i$  and  $R_j$  are the radii of the atoms in Equation 2.2.

Every atom pair interacts through a steric interaction given by the first three terms of Equation 2.3. Additionally, depending on the atom type, there may be hydrophobic and non-directional Hydrogen Bonding interactions, as indicated by the last two terms.

$$e_{\text{pair}}(d) = w_1 \times \text{Gauss}_1(d) + w_2 \times \text{Gauss}_2(d) \\ + w_3 \times \text{Repulsion}(d) + w_4 \times \text{Hydrophobic}(d) + w_5 \times \text{HBond}(d) \quad (2.3)$$

Steric interaction in Vina is assessed using three terms (Equations (2.4) to (2.6)). The combination of an attractive Gaussian function with a repulsive parabolic function reproduces the general shape of a typical Lennard-Jones interaction, provided the Gaussian term is negative and the parabolic positive. If both atoms in the pair are hydrophobic, the linear function in Equation 2.7 is included. Additionally, if the pair consists of an H-bond donor and an H-bond acceptor, Equation 2.8 is also applied. These last two equations are simple, piecewise linear, and their effect can be thought of as modifying the steric interaction to produce an increased attraction when these types of interactions are present.<sup>18</sup>

$$\text{Gauss}_1 = e^{-\alpha r_{ij}^2} \quad (2.4)$$

$$\text{Gauss}_2 = e^{-\beta(r_{ij}-r_o)^2} \quad (2.5)$$

$$\text{Repulsion}(d) = \begin{cases} d^2 & \text{for } d \leq 0 \\ 0 & \text{for } d > 0 \end{cases} \quad (2.6)$$

$$Hydrophobic(d) = \begin{cases} 1 & \text{for } d \leq p_1 \\ p_2 - d & \text{for } p_1 > d < p_2 \\ 0 & \text{for } d \geq p_2 \end{cases} \quad (2.7)$$

$$HBond(d) = \begin{cases} 1 & \text{for } d \leq h_1 \\ \frac{d}{-h_1} & \text{for } h_1 > d > 0 \\ 0 & \text{for } d \geq p_2 \end{cases} \quad (2.8)$$

$$\begin{aligned} \Delta G_{bind} = & \sum_{ij} w_1 \times e^{-\alpha r_{ij}^2} + \sum_{ij} w_2 \times e^{-\beta (r_{ij} - r_o)^2} + \sum_{ij} w_3 \times S_{repulsion}(r_{ij}) \\ & + \sum_{ij} w_4 \times S_{hydrophobic}(r_{ij}) + \sum_{ij} w_5 \times S_{hydrogen\ bond}(r_{ij}) + \Delta G_{torsion} \end{aligned} \quad (2.9)$$

This computational efficiency is balanced by a scoring function that approximates key interaction types with reasonable physical realism, thereby avoiding the need for full quantum calculations. AutoDock Vina's search algorithm is based on an iterated local search global optimizer that combines global exploration with local optimization. The process begins with a random sampling of ligand conformations and poses within the predefined search space (binding site). Then, the scoring function scores each candidate pose. The pose is iteratively perturbed in translation, rotation, or torsion across a local optimization of the conformation and accompanied by a search for lower energy states. This hybrid strategy may prevent the local minima and may increase the chance of finding the global minimum, which is the correct binding pose.

Mathematically, the algorithm optimizes the function:

$$\min_{x \in \Omega} \Delta G_{bind}(x)$$

where  $x$  represents the vector of ligand position, orientation, and torsion angles, and  $\Omega$  defines the search space constrained by the docking box dimensions.

By automating the docking process, the Vina Dock Tool significantly reduces the manual effort involved, streamlining the workflow for drug discovery and molecular interaction studies. The user provides input files containing the ligand and receptor structures, sets docking parameters, and specifies options for the docking site. The tool then runs the simulation, generates output files, and provides results that help identify the most promising ligand binding poses.

Let's start the docking process in the Galaxy BRIDGE tool<sup>19,20</sup>. The user should provide two input files, which must be a ligand PDBQT file and a receptor PDBQT file (the user can prepare these files using the Galaxy BRIDGE tools). The ligand PDBQT file is a file containing the ligand's 3D structure, a small molecule being screened for its ability to bind to the target protein. The ligand is in PDBQT format, containing the 3D conformations of each ligand atom, as well as the atomic type and partial atomic charges of each atom. It is essential to have such a format, as AutoDock Vina requires it to simulate the interaction between the ligand and the protein, thereby providing reliable docking results. The receptor file contains the 3D structure, most likely in PDBQT format, which is the target molecule connected to the ligand. The protein file contains information about the receptor's structure, including the binding site to which the ligand is to be docked. Altogether, these files set up the molecular environment for the docking simulation.

After uploading the ligand and receptor files, the user should specify several key docking parameters. The first parameter specifies the number of poses to generate. This number represents the number of different binding poses into which each ligand should be docked. Since ligands can bind to the protein in different ways, generating a specified number of poses allows the tool to assess various potential binding modes. The default number of poses is 10, and the user can modify this based on the detail required for the study. The greater the number of poses, the more complete the search, but it will also increase computational time.

The second is how exhaustive the docking should be. Exhaustiveness refers to the thoroughness employed by the docking algorithm in its search for the optimal pose binding. A greater exhaustiveness value results in a more exhaustive binding site search, with a better chance of finding the optimal binding pose. But it also means longer computation time. For default exhaustiveness (32), otherwise, the value can be changed according to the users' requirements and computational availability.

Along with these principal settings, the tool contains two optional settings for defining the docking site, providing a bound ligand, or specifying coordinates. If the user selects providing a bound ligand, they need to upload a file containing a bound ligand in PDB, MOL2, or SDF format. This is relevant because the docking algorithm can specify the docking site more accurately when it is in the presence of a bound ligand. The docked ligand file shows the docking tool's view of the ligand's interaction with the protein, which is used to direct the docking search.

Instead, the user can select to provide coordinates in which they enter the center of the docking box (X, Y, Z). This option provides flexibility when no bound ligand is available. After entering the center coordinates, the user must provide the box size (X, Y, Z) in Angstroms (Å),

which determines the search region for the docked ligand. This step is rather important for restricting the volume of the protein to be explored for docking, such that the search is concentrated at that part of the protein.

After the inputs and parameters are configured, the user presses the execute button, and the docking simulation begins. The tool implemented by AutoDock Vina performs the docking process, searching for the optimal poses of the ligand within the specified docking box. In the simulation, the docking algorithm evaluates the potential interactions between the ligand and the receptor, scoring each pose based on its predicted binding affinity. The results are saved in output files, which consist of the docked poses PDB file, the Results CSV file, and the log TXT file.

The docked poses PDB file contains the final docking poses, each docking pose representing a different binding orientation of the ligand within the protein. These poses are saved in PDB format, which can be visualized using molecular graphics tools such as PyMOL or Chimera to examine the ligand-protein interactions in depth. The Results CSV file contains the docking scores for the generated poses in CSV format, with lower scores indicating stronger binding. These scores are used to select the top docking poses that are worth further examination. Finally, the log TXT output file includes a record of the docking simulation, providing detailed information about the docking process, any errors encountered, and the computational steps taken.

The Tool is a useful resource for researchers in drug discovery and molecular biology, as it enables the automation of the docking process, simplifying the task of screening large compound libraries. The tool accepts ligand database files provided in SDF format. It can dock multiple ligands in a single run, making it an efficient and scalable solution for virtual

screening. The ability to quickly generate docking poses and scores enables researchers to identify promising candidates for further experimental validation, significantly accelerating the drug development process. Next, explore the tool’s structure in detail by examining the Python script (docking.py) alongside its XML configuration file.

### 3.2 Python Script(docking.py)

This Python script has been created for the Galaxy BRIDGE tool as an interface that connects external computational tools or scripts with the Galaxy platform. It enables seamless integration into Galaxy workflows and typically consists of components for parsing input parameters, executing backend commands or scripts, and formatting outputs to ensure compatibility with the Galaxy framework. This allows users to execute complex bioinformatics (or cheminformatics) tasks through Galaxy’s web-based environment while leveraging the functionality of custom or third-party tools managed by a Python script that streamlines the data flow and maintains reproducibility.

---

```
import argparse
from rdkit import Chem
from rdkit.Chem.rdMolTransforms import ComputeCentroid
from vina import Vina
import os
import numpy as np
from multiprocessing import Pool, cpu_count
```

---

In this section imports all the necessary Python modules and functions the script needs. First, `import argparse` imports a built-in module that enables the program to read and handle command-line arguments, then users can customize inputs when running the tool. Next, the chemical toolkit and RDKit’s<sup>21</sup> chemical toolkit will be imported `from rdkit import Chem` line, which is important for loading and manipulating molecular structures. Then, `from`

`rdkit.Chem.rdMolTransforms` `import ComputeCentroid` imports an important RDKit function that calculates the 3D geometric center (centroid) of a molecule very useful for finding where the ligand is positioned in space. Run molecular docking with AutoDock Vina `from vina` `import Vina` line, will directly import the Vina docking engine, and the script will be able to interface with AutoDock Vina. The `import os` line simply puts the import Python base library to interact with the running operating system, such as making directories or dealing to file paths. The script also imports NumPy<sup>22</sup> with `import numpy as np`, and this library is powerful for fast calculations, especially for processing data as arrays, including docking score. Finally, `from multiprocessing import Pool, cpu_count` `import tools` to run multiple jobs in parallel using a pool of worker processes equal to the number of CPU cores, the above-listed number of CPU cores. it accelerated the entire docking process for the larger ligand dataset to verify. Altogether, these imports load the script with everything for chemistry, docking, file handling, and parallel execution.



---

```
def parse_arguments():
    parser = argparse.ArgumentParser(description="Dock ligands to a protein using
AutoDock Vina and RDKit")
    parser.add_argument('ligand1', type=str, help="Path to the ligand file in SDF
format (can contain multiple ligands)")
    parser.add_argument('ligand2', type=str, help="Path to the second ligand in
PDBQT format")
    parser.add_argument('protein', type=str, help="Path to the protein in PDBQT
format")
    parser.add_argument('output', type=str, default="Ligand_vina_out", help="Output
folder for docked poses")
    parser.add_argument('n_poses', type=int, default=10, help="Number of poses to
dock (default: 10)")
    parser.add_argument('box_size', type=float, nargs=3, default=[20, 20, 20],
                        help="Box size for docking as height, width, and length
(default: 20, 20, 20)")
    parser.add_argument('exhaustiveness', type=int, default=32,
                        help="Exhaustiveness of docking (default: 32)")
    parser.add_argument('score_output', type=str, default="docking_scores.txt",
                        help="File to save docking scores")
    parser.add_argument('max_scores', type=int, default=10, help="Maximum number of
docking scores to select ")
    return parser.parse_args()
```

---

Python's `argparse` module is a powerful library which helps us to parse command-line arguments quickly but without hardcoding the data inside the script. The first thing the function does is create a new `ArgumentParser` object (called `parser`) from the `argparse` module. This parser was enabled with a description string such as: "The script docks ligands to a protein with AutoDock Vina and RDKit" It also makes it easier for users to see why your script is implemented when they get help. Then the function takes multiple arguments in order from the user. This value is represented by the first argument, 'ligand1' and needs to be a string and is the file path only to a single ligands' file in SDF format potentially flooded with a numerous amount of ligands as can be seen in the help message. 'ligand2' parameter is a string input which requires a path to a second ligand file in pdbqt file format, and they are necessary for the docking. The third parameter, 'protein,' is the route to the protein receptor of the ligand (again in PDBQT). Then, the 'output' argument is appended to specify the folder to store docked poses files, by default this is "Ligand\_vina\_out" default value not if you set it

yourself. After that, `n_poses` is an integer argument setting the number of docking poses generated for each ligand, and the default is 10. The next argument, 'box\_size', is a little different in that it takes three float values (containing height, width, and length, respectively), and this being a special case, we pass `nargs=3` and `[20,20,20]` to force this argument. Then, `exhaustiveness` is an integer that specifies how exhaustively the docking algorithm should search for binding poses; 32 is used as the default and is a tradeoff between computation time and precision. The 'score\_output' argument is a string specifying the name of the file in which to store the docking scores (default: "docking\_scores. txt". Lastly, the argument 'max\_scores' specifies how many top docking scores to preserve in the output; by default, 10. The parser closes the function with `parser. parse_args()` reads the command-line arguments, validates and converts as necessary to match the definitions given, applies any default values given, and returns them as an object whose members are the input values. All user inputs are available for easier manipulation, controlling the docking pipeline interactively later in the program.

---

```

def dock_ligand(ligand_info):
    ligand_mol, ligand_id, ligand2_path, protein_path, box_size, exhaustiveness,
    n_poses, output_folder = ligand_info

    try:
        v = Vina(sf_name='vina')
        with open(protein_path, 'r') as protein_file:
            protein_content = protein_file.read()
        v.set_receptor(protein_path)
        centroid = ComputeCentroid(ligand_mol.GetConformer())
        v.set_ligand_from_file(ligand2_path)
        v.compute_vina_maps(center=[centroid.x, centroid.y, centroid.z],
box_size=box_size)
        v.dock(exhaustiveness=exhaustiveness, n_poses=n_poses)
        combined_file = os.path.join(output_folder, f"{ligand_id}.pdbqt")
        with open(combined_file, 'w') as output_combined:
            docked_poses = v.poses(n_poses=n_poses).split("ENDMDL")
            for pose_num, pose in enumerate(docked_poses):
                if pose.strip():
                    output_combined.write(f"MODEL {pose_num + 1}\n")
                    output_combined.write(protein_content + "\n")
                    output_combined.write(pose.strip() + "\nENDMDL\n")

            docking_scores = v.energies()
            docking_scores = np.array(docking_scores)
            result_scores = []

            if docking_scores.size > 0:
                for pose_num, score in enumerate(docking_scores):
                    if score.size > 0:
                        pose_id = f"{ligand_id}_pose_{pose_num + 1}"
                        first_score = score[0]
                        result_scores.append((pose_id, first_score))

            return result_scores

    except Exception as e:
        print(f"Error docking ligand {ligand_id}: ")
        return []

```

---

The `dock_ligand` function first unpacks all input parameters from `ligand_info` (input tuple), which consists of the ligand molecule object, the ligand id, paths for the second ligand and protein files, docking box size, exhaustiveness, number of poses to generate, and the output folder where the docked poses will be saved. It then encases its docking process within a try-except block to prevent crashing the entire program in the event of errors. First, within the try block, it prepares a new Vina object, which will handle the docking and scoring. It then opens the protein file in read mode and stores the full content in a string variable, as it will be useful to merge the docked ligand poses with the receptor structure later. It's next to assign the

receptor structure to the Vina object using the path to the protein file. Next, it calculates the centroid of the ligand 3D conformation by invoking RDKit's `ComputeCentroid` function based on the ligand's conformer, which is the average position of the ligand atoms and will be used as the center of the docking search box. The second ligand (identified by its PDBQT file path) is loaded into Vina for docking against the receptor. After that, the function tells Vina to calculate the docking maps for the box of size `box dimension` and with the box centered on the calculated centroid (this space is where the docking will take place), as described here: The actual docking process is initiated by running `v.dock()` with the exhaustiveness, number of poses parameters, thus telling Vina to look for the best ligand binding poses. When docking is complete, the script is ready to save the results, which it does by establishing the output path based on the ligand ID and the output directory. It then opens this file in write mode, reads the Vina docked poses, and extracts individual docked poses, separated using the PDBQT standard model delimiter "ENDMDL". For each pose, excluding empty ones, it loops and writes a MODEL header with the pose number, copies the protein structure content, writes the ligand pose, and concludes with an ENDMDL tag. It retrieves the docking energy scores for each pose and saves them as a NumPy array. It initializes an empty list to store the results and iterates over all the pose scores. It creates a unique pose ID for each valid scoring by combining the ligand ID and the pose position number, and extracts the first value of DC-OPLS energies (i.e., the binding affinity). This tuple is then appended to the results. At last, the `pose_ids` and the corresponding `docking_scores` are the returned lists of the function. If an error occurs during this process, the except block catches the exception, prints an error message containing the ligand ID and a detailed exception, and returns an empty list, allowing the rest of the docking workflow to proceed without interruption.

---

```

def main():
    args = parse_arguments()

    if not os.path.exists(args.output):
        os.makedirs(args.output)

    ligands = Chem.SDMolSupplier(args.ligand1, removeHs=False)
    if ligands is None or len(ligands) == 0:
        raise ValueError("No ligands found in the SDF file.")

    ligand_info = []
    for i, lig in enumerate(ligands):
        if lig is None:
            print(f"Skipping invalid ligand at index .")
            continue

        ligand_id = lig.GetProp('_Name') if lig.HasProp('_Name') else f"ligand_"
        ligand_info.append((lig, ligand_id, args.ligand2, args.protein,
args.box_size, args.exhaustiveness, args.n_poses, args.output))

    with Pool(processes=cpu_count()) as pool:
        all_results = pool.map(dock_ligand, ligand_info)

    all_scores = [score for sublist in all_results for score in sublist]
    all_scores.sort(key=lambda x: x[1])
    top_scores = all_scores[:args.max_scores]

    with open(args.score_output, 'w') as score_file:
        for pose_id, score in top_scores:
            score_file.write(f"{pose_id}\t{score:.3f}\n")
            print(f"{pose_id} docking score: {score:.3f}")

if __name__ == '__main__':
    main()

```

---

The `main()` function begins by invoking `parse_arguments()` to extract all user inputs and parameters from the command line. The script will need files and options. Then, it checks to see if the output directory I specified exists. if it doesn't, it creates this directory so the docking result can be stored somewhere. Second, it reads in all the ligands from the input SDF file through RDKit's `SDMolSupplier`, which allows multiple molecules per file – if no ligands are found, or if the file is empty, the function will raise an error and stop running. The script then creates a list `ligand_info` to store all of the information necessary to dock each ligand. This comprises the ligand molecule object, its ID (extracted from the ligand's Name property or a fallback name), and all of the docking parameters specified as arguments. It passes over

any invalid ligands encountered during loading with a warning. With this list prepared, the function constructs a multiprocessing pool with the number of worker processes set to the number of CPU cores, allowing many ligands to be docked simultaneously to speed up the process. Using the `pool.map()` here, which spreads the docking tasks out by applying the `dock_ligand` function to each item of `ligand_info` and collecting all the parallel docking results. After all these individual docking scores, one for each structure result, are merged into a list per ligand, the code iterates through those structures and flattens the list to a single list of pose-score pairs. It then orders this list by docking score, best (lowest) to worst (highest), and only retains the top `N` values, where `N = max_scores` (a parameter set by the user). Lastly, it outputs the top docking scores and their pose IDs to the output score file by writing the scores to three decimal places and echoing each score to the screen for real-time feedback. The main body of the script is at the end of the file, triggered by the standard Python if `__name__ == 'main':` that is used to check whether the `main()` function should be executed. The test works by checking whether the script is being run as a standalone script and not being imported as a module. This configuration enables us to encapsulate the entire docking pipeline, from input parsing and ligand loading to multicore docking and output collection and writing, in a neat, easy-to-use, and efficient workflow.

This concludes the general description of the main docking workflow, encompassing argument parsing and ligand preparation, parallel docking execution, and result manipulation. If you dig a bit deeper into how the docking configurations and parameters are handled internally, you will need to take a closer look at the XML configuration. This XML script serves as a protocol file containing the exact terms of the docking process, such as the dimensions of the search space, levels of exhaustiveness, or criteria for generating output. In the following

section, we will discuss the format and contents of this XML file and demonstrate how it interacts with the docking engine to efficiently design and automate simulations for the docking of the molecules to be modeled.

### 3.3 XML Wrapper File

This Galaxy XML wrapper is a configuration file that declares how a tool is wrapped within the Galaxy framework. It details the interface of the tool (its inputs and outputs, what the parameters are, how you run it with the command line and how the results are visualized). The wrapper instructs Galaxy on how to display the tool to end-users, how to invoke it based on the underlying script or program (written in Python or other languages), and what to do with the data before and after execution. This XML file works as an interface between Galaxy's user view and the tool's backend.

---

```
<tool id="docking_tool" name="Molecular_Docking" version="@TOOL_VERSION@+galaxy@GALAXY_VERSION@">
```

---

This line defines the root element of the XML file `<tool>`, which represents a single tool configuration in a Galaxy workflow or similar system. It takes some key attributes: `id="docking_tool"`. This `id` is how the tool can be referred to (internally) elsewhere. The `name="Molecular_Docking"` attribute is the title of the tool that the user will actually see in their UI. The `version="@TOOL_VERSION@+galaxy@GALAXY_VERSION@"` indicates the tool's version, but it uses variables (`@TOOL_VERSION@` and `@GALAXY_VERSION@`) that will be replaced at runtime, likely at deployment time, with the version of the tool and the version of the Galaxy platform on which it is running. This will help tracking support and making sure the tool is consistent with different versions.

---

```
<macros>
<token name="@TOOL_VERSION@">1.5.7</token>
<token name="@GALAXY_VERSION@">0</token>
</macros>
```

---

This snippet defines a `<macros>` section in the XML, which is used to set reusable tokens or placeholders throughout the tool definition. Here, two `<token>` elements assign specific values: `@TOOL_VERSION@` is set to "1.5.7", indicating the version of the docking tool itself, and `@GALAXY_VERSION@` is set to "0", likely representing the version of the Galaxy platform or a placeholder for it. These tokens will substitute for the original placeholders used in the `version` attribute for the root element and allow the version in the single place to be updated easily without modifying every occurrence of the version in the file.

---

```
<requirements>
<requirement type="package" version="@TOOL_VERSION@">mgltools</requirement>
<requirement type="package" version="2022.09.5">rdkit</requirement>
<requirement type="package" version="1.21.2">numpy</requirement>
<requirement type="package" version="1.2.3">vina</requirement>
<requirement type="package">multiprocessing</requirement>
</requirements>
```

---

This section describes the software dependencies needed for the tool to run properly. The `<requirements>` tag groups multiple requirement entries, each declaring a required package. Each `<requirement>` has a `type="package"` attribute to indicate that these are software libraries or modules that need to be installed. The first requirement is `mgltools` with a version placeholder `@TOOL_VERSION@`, meaning it should match the docking tool's version defined earlier. Next are specific versions of popular Python libraries: `rdkit` version `2022.09.5` for chemical informatics, `numpy` version `1.21.2` for numerical operations, and `vina` version `1.2.3`, which provides the docking functionality. Finally, `multiprocessing` does not have a version because it's a standard Python library, which is part of the core deliverable set. This is so that the Galaxy



environment or any system that runs this tool can have these dependencies updated in order to be compatible and reproducible.

---

```
<stdio>
<exit_code range="1"/>
</stdio>
```

---

This section instructs the tool on how to interpret its exit status codes when it is inside the block. The component declares that, in the event the tool's process fails with an exit status of 1, the Galaxy system will also consider it an error. Exit codes are numeric values that one program can pass to the system to indicate how it finished, whether in success or failure. 0 is the conventional code for a successful run, and non-zero values, such as 1, are often employed to convey an error condition. By stating this explicitly, the tool integration enables Galaxy to correctly identify when the docking tool crashes while running and to act accordingly, which may involve warning the user that something has gone wrong or aborting the workflow.

---

```
<command><![CDATA[
    cp $__tool_directory__/docking.py . &&

    ln -s $ligand2 ./ligand2.mol2 && prepare_ligand4.py -l ./ligand2.mol2 -v -o
ligand2.pdbqt &&
    ln -s $protein ./protein.pdb && prepare_receptor4.py -r ./protein.pdb -v -o
protein.pdbqt &&
    ln -s $ligand1 ./ligand1.sdf &&

    python docking.py ligand1.sdf ligand2.pdbqt protein.pdbqt output_folder
$poses $box_size_x $box_size_y $box_size_z $exhaustiveness docking_scores
$number_of_docking_scores &&

    python convert.py ./output_folder ./docked_ligand_porses &&

    tar cf docked_ligand_porses.tar docked_ligand_porses/ &&
    cp docked_ligand_porses.tar $docked_ligand_porses &&
    cp docking_scores $docking_scores

]]></command>
```

---

The first command `cp` copies the main dockings Python script, `docking.py` located in the installation directory `$_tool_directory_` into the current working directory. This will make the script available locally to the host. Next, `ln -s $ligand2 ./ligand2.mol2` creates symbolic link `ligand2.mol2` is formed using the user's second ligand file `$ligand2`(python script input ligand in sdf format). This symbolic link standardizes the filename so the following commands can always refer to `ligand2.mol2`, no matter what the original filename is. The fourth command is the `prepare_ligand4.py -l ./ligand2.mol2 -v -o ligand2.pdbqt` launches the `prepare_ligand4.py` script on this linked ligand file, which converts and prepares the file into the PDBQT format (`ligand2.pdbqt`) required by AutoDock Vina; the `-v` flag enables verbose output for detailed logs during processing. After that, `ln -s $protein ./protein.pdb` creates a symbolic link named `protein.pdb` will be written for the user's protein input file (`$protein`), which is also used to normalize the filename for the prep step and remove any extensions. The sixth command is `prepare_receptor4.py -r ./protein.pdb -v -o protein.pdbqt` calls `prepare_receptor4.py` to handle the linked protein file and convert it to PDBQT format (`protein.pdbqt`) for docking, and verbose logging. Then, `ln -s $ligand1 ./ligand1.sdf` a symbolic link called `ligand1.sdf` is created pointing to the ligand input file (`$ligand1`) in the Python script, which can contain multiple ligands for docking. Following input preparations, the main docking script, `docking.py`, is executed with `python docking.py ligand1.sdf ligand2.pdbqt protein.pdbqt output_folder $poses $box_size_x $box_size_y $box_size_z $exhaustiveness docking_scores $number_of_docking_scores` command and all necessary inputs and parameters, including the ligand and protein files in PDBQT and SDF formats, the output folder name (output), the number of poses (`$poses`), the box sizes (`$x`, `$y`, `$z`), the level of exhaustiveness (`$exhaustiveness`), and output files for docking scores (`scores`, `$docking_scores`). Once docking finishes, the docking output files from the `./output` folder into

a directory called `./docked_ligand_poses.tar` of `docked_ligand_poses.tar` `docked_ligand_poses`. This command uses the `zip` function and compresses the `docked_poses` folder into a tar folder called `docked_ligand_poses.tar` using the `tar cf` command. The tar archive is then copied to the specified Galaxy output location `$docked_ligand_poses`, making it accessible as a dataset within the Galaxy platform. Lastly, `cp docking_scores $docking_scores`, the file containing the docking scores, to the directory where the docking scores text file was expected to be copied to `$docking_scores` for the user to examine the results or to analyze them further. This well-ordered sequence of commands can be executed to completely automate the input preparation, docking, result conversion, and packaging in the Galaxy environment.

---

```
<inputs>
  <param type="data" name="protein" format="pdb" label="protein PDB" />
  <param type="data" name="ligand2" format="mol2" label="Reference ligand" />
  <param type="data" name="ligand1" format="sdf" label="Ligand to Dock" />
  <param type="text" name="poses" format="txt" label="Number of poses to output" value="10"/>
  <param type="text" name="exhaustiveness" format="txt" label="exhaustiveness" value="32"/>
  <param type="text" name="box_size_x" label="Box size X (Å)" value="20" />
  <param type="text" name="box_size_y" label="Box size Y (Å)" value="20" />
  <param type="text" name="box_size_z" label="Box size Z (Å)" value="20" />
  <param type="text" name="number_of_docking_scores" format="txt" label="Number of top docking poses" />
</inputs>
```

---

This `<inputs>` section specifies all parameters and data files that the user must provide to run the docking process. Each `<param>` element specifies an input with attributes that describe its type, name, expected file format, label shown to the user, and default values if applicable. The first three inputs are data files: the `protein` parameter expects a PDB file format

containing the receptor protein structure; `ligand2` is the reference ligand in MOL2 format, often used as a known binder for comparison; and `ligand1` is the ligand or set of ligands to be docked, expected in SDF format, which may contain multiple molecules. Below are a few textual parameters you may use to control docking: `poses` specify the number of docking poses the script needs to produce (default=10); `exhaustiveness` determines how thorough the docking search should be (default=32). Next, the `docking_box_size` is broken down into `box_size_x`, `box_size_y` and `box_size_z`, 20 angstroms by default, and they set the region in space (in 3D connected to the ligand), where docking is done. Finally, `number_of_docking_scores` permits the user to indicate how many of the top docking scores he wishes to keep, and output. These inputs, combined, enable users to control the files and parameters required for performing docking, and as a result, make the tool flexible and interactive.

---

```
<outputs>
  <data name="docked_ligand_poses" format="tar" />
  <data name="docking_scores" format="txt"/>
</outputs>
```

---

This `<outputs>` section describes the files that the tool will generate and return at the end of its run. Each `<data>` element identifies an output dataset by giving it a `name` and a `format`. The first output, named `"docked_ligand_poses"` (probably `"docked_ligand_poses"`), is a tar archive file (`format="tar"`), which contains all the docked ligand structures for easy download and further analysis. The second output, `"docking_scores"`, is a plain text file (`format="txt"`) that contains the numerical docking scores for the ligand poses, summarizing how well each ligand binds to the protein. These well-defined outputs mean that once docking is finished, the Galaxy platform knows which files to expect, indicates if they are present, and exposes them to the user or downstream workflow steps.

---

```
<help>
```

```
</help>
```

---

This `<help>` section provides detailed manual information written directly in the tool's XML definition. The documentation has been created to instruct the user on how to use the tool, what to provide as input, and what to expect as output, thereby promoting clarity in the tool's use within the Galaxy environment.

---

```
< Citations>
```

```
< citation type="doi">10.1002/jcc.21334</ citation>
```

```
</ Citations>
```

---

This `< Citations>` section provides academic references to the tool, helping users understand its scientific basis or methodology. Inside, the `< citation>` element specifies a reference in the form `type="doi"`, which is the abbreviation for Digital Object Identifier and is a machine-readable form of a persistent link to a paper. The DOI 10.1002/jcc.21334 is an article, probably presenting the AutoDock Vina algorithm as well as related molecular docking techniques that the tool relies upon. Citing in this way recognizes the original authors while allowing the user to find more supporting background for the model and endorses the scientific integrity of the tool's approach.

### 3.4 Benchmarking Galaxy BRIDGE Tool: RMSD-Based Evaluation.

After completing the docking simulation, the RMSD was calculated by evaluating the precision of the Galaxy BRIDGE tool. RMSD is the quantitative estimation of the deviation of the predicted ligand pose relative to the experimental one. RMSD was calculated with PyMOL<sup>23</sup>, a molecular visualization tool. The predicted ligand structure can be aligned with

the experimental one using PyMOL, and the distance between equivalent atoms in their respective conformations is calculated. RMSD value is calculated using Equation (2.10)

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left( x_i^{(pred)} - x_i^{(exp)} \right)^2} \quad (2.10)$$

Where N is the number of atoms in the ligands being compared,  $x_i^{(pred)}$  and  $x_i^{(exp)}$  are the coordinates of the predicted and experimental ligand pose, respectively. It determines average squared differences between the atomic positions in the predicted and experimental ligand poses, providing a quantitative measure of docking accuracy. The lower the RMSD value indicates the better the performance of the docking prediction. This benchmark evaluates the performance of the galaxy BRIDGE tool using the COACH benchmark dataset, sourced from the Zhang Group's website (<https://zhanggroup.org/COACH/benchmark>), which provides experimentally validated binding site annotations derived from high-quality structural data. In this study, “Training set” was chosen. The dataset included several receptor-ligand complexes, each with a known experimental structure, which serves as the reference for assessing the docking performance. These structures were used as references for the docking process. The docking simulations were conducted using the Galaxy BRIDGE tool, and both the receptor and ligand adjusted their conformations to find the most stable binding pose. The Galaxy BRIDGE tool generates multiple poses (selected by the user) for each receptor-ligand complex and selects the pose with the best predicted binding energy for further analysis. The best pose was selected and compared with the COACH benchmark dataset.

### 3.5 Result and Discussion.

To determine the binding affinity and structural stability of ligand-protein interactions, a comparative analysis of the standard and docked binding residues across 50 protein-ligand complexes directed against the dengue virus. The binding site residues for both the crystallized and docked complexes were identified and tabulated (Table 2.2), and RMSD values were computed to check the conformational changes upon docking. This analysis provides insight into the accuracy of the docking procedure and the consistency of binding interactions.

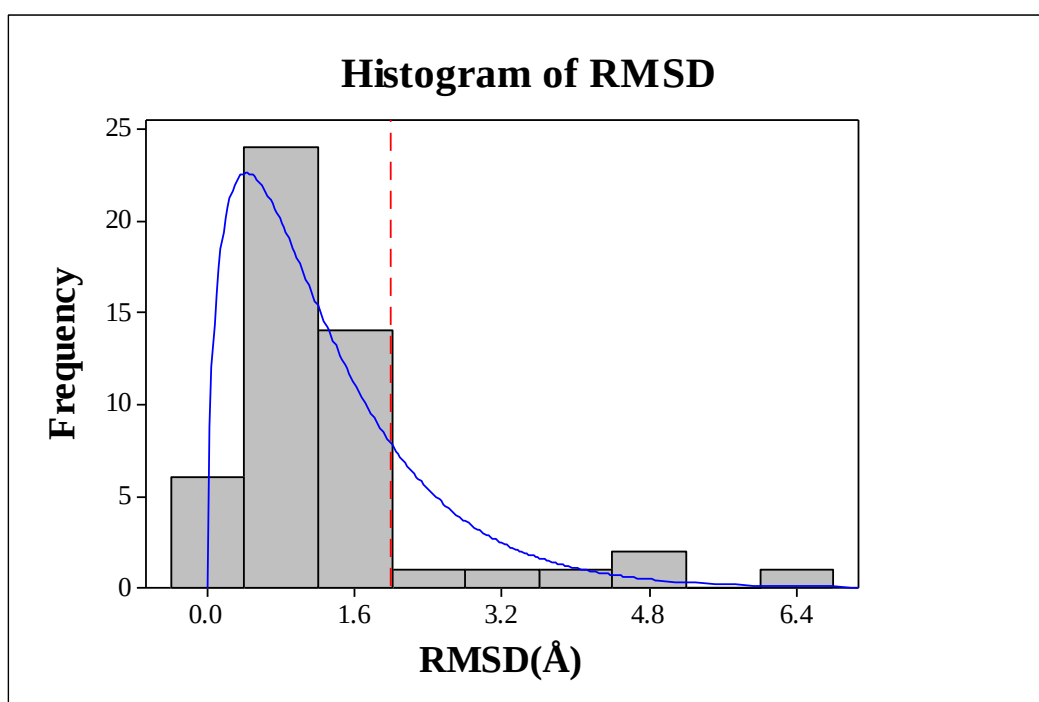
**Table 2.2.** Comparison of standard and docked binding residues for various protein-ligand complexes, along with their corresponding Root Mean Square Deviation (RMSD) values indicating docking accuracy.

| No | Protein-Ligand  | Standard Binding Residues                                                | Docked Binding Residues                                    | RMSD  |
|----|-----------------|--------------------------------------------------------------------------|------------------------------------------------------------|-------|
| 1  | 1abeA_BS01_UUU  | 9,13,15,89,107,146,150,203,204,231                                       | 9,89,150,203                                               | 0.044 |
| 2  | 1af7A_BS01_SAH  | 28,80,81,82,83,84,88,113,115,119,144,145,146,201,202,203,220,221,222,226 | 28,37,91,93,121,127,153,154,211,226,227                    | 0.633 |
| 3  | 1ag9B_BS04_BTBT | 147,149,150                                                              | 123,125,147,149                                            | 0.771 |
| 4  | 1akuA_BS01_FMN  | 9,10,11,12,13,14,57,58,59,60,92,93,94,97,99,100,101                      | 11,12,14,15,57,58,59,60,61, 92,,94,99,101                  | 0.333 |
| 5  | 1b9hA_BS01_PLP  | 57,58,59,62,83,84,86,155,157,158,179,181,184                             | 57,58,59,83,84,155,157,181,184                             | 1.659 |
| 6  | 1bgvA_BS01_GLU  | 89,91,110,113,125,163,164,165,377,380                                    | 91,110,125,163,165,380                                     | 1.217 |
| 7  | 1btnA_BS01_I3P  | 8,21,22,23,69,71                                                         | 8,23,69,71                                                 | 1.246 |
| 8  | 1bx4A_BS02_ADN  | 11,13,15,37,61,62,65,120,167,293,297                                     | 13,17,64,65,67,293,297                                     | 0.53  |
| 9  | 1bx4A_BS03_ADN  | 262, 264, 265, 281, 283, 286, 289, 295, 296, 321, 324, 328               | 262, 264, 265, 281, 283, 286, 289, 295, 296, 321, 324, 328 | 0.256 |
| 10 | 1cqxB_BS01_HEM  | 42,43,44,80,81,84,85,88,90,94,95,98,126,129,130,133                      | 42,43,80,81,84,85,88,90,94,98,126,129,130,133              | 0.746 |
| 11 | 1eyqA_BS01_NAR  | 33,35,45,47,98,103,106,107,110,187,188                                   | 98, 103, 106, 107, 110, 187, 188                           | 0.421 |
| 12 | 1f2dC_BS01_PLP  | 50,51,54,79,200,201,202,203,205,295,296,323,324,325                      | 50,54,79,200,201,202,203,205                               | 0.69  |
| 13 | 1f2oA_BS04_LEU  | 85,97,131,132,160,161,199,202,219,246,247                                | 85,97,132,160,161,171,246,247                              | 1.285 |
| 14 | 1fihB_BS01_NGA  | 113, 115, 117, 126, 138, 139                                             | 201,209,211,215                                            | 0.589 |
| 15 | 1g38D_BS03_NEA  | 1,27,51,52,56,68,69,70,85,86,87,126                                      | 1,27,51,52,56,68,69,70,85,86,87,126                        | 1.2   |
| 16 | 1g55A_BS01_SAH  | 9,10,11,12,14,33,34,35,54,55,56,57,75,77,79,297,298                      | 11,12,13,34,54,57,75,77,79,297,298                         | 0.571 |
| 17 | 1g6iA_BS01_UUU  | 63,118,476                                                               | 39,63,95,96,118,149,476,513,514,516,519                    | 4.805 |
| 18 | 1g6iA_BS03_DMJ  | 305,394,395,396, 460, 464, 486, 487                                      | 305,394,412,413,486,487                                    | 0.056 |
| 19 | 1gd9A_BS01_PLP  | 94,95,96,120,166,170,198,200,201,231,232,240                             | 95,96,120,166,170,198,200,201,231,232,240                  | 1.018 |

|    |                |                                                                                     |                                                                     |       |
|----|----------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------|
| 20 | 1h69B_BS02_ARH | 105, 106, 149, 150, 154, 161, 194                                                   | 149,150,154,                                                        | 0.674 |
| 21 | 1h69B_BS04_ARH | 68,128,178                                                                          | 68,128,178                                                          | 1.089 |
| 22 | 1hf3B_BS03_NAD | 46,47,48,51,174,178,199,201,202,203,223,228,268,269,292,293,294,317,318,319,362,369 | 46,47,48,174,178,202,223,228,268,292,294,317,318,369                | 0.628 |
| 23 | 1hl7B_BS01_BD1 | 31,32,97,98,99,125,162,166,204,232,233,259                                          | 31,32,97,98,99,125,204,232,233,259                                  | 0.674 |
| 24 | 1i7qB_BS01_GLU | 55,56,57,58,83,84,87,131,132,133,134                                                | 55,56,57,134,151                                                    | 1.36  |
| 25 | 1icuB_BS02_NIO | 70,166                                                                              | 70                                                                  | 0.64  |
| 26 | 1icuB_BS03_NIO | 40,41,124                                                                           | 40,41,124                                                           | 0.129 |
| 27 | 1j20A_BS01_AMP | 6,13,31,32,33,37,92,113,114,125,170                                                 | 6,13,31,32,34,38,92,113,114,125,170                                 | 0.497 |
| 28 | 1j20A_BS02_AS1 | 84,88,89,115,116,119,120,121,124,177,179,253,265,305                                | 8488, 89,120,121,124,177,179,253,265,305                            | 0.879 |
| 29 | 1jbwA_BS03_TMF | 15,74,75,82,120,121,399,402                                                         | 15,74,75,82,120,121,399,402                                         | 0.723 |
| 30 | 1jk7A_BS03_OKA | 91, 120, 125, 129, 215, 216, 267, 268, 270, 271                                     | 91, 120, 216, 267, 268 ,270, 271                                    | 1.385 |
| 31 | 1jrza_BS01_HEM | 8,9,14,17,18,24,69,73,74,75                                                         | 8,9,14,69,71,74                                                     | 1.293 |
| 32 | 1jrza_BS06_FUM | 170, 365, 375, 377, 378, 402, 504, 544, 546, 547                                    | 170, 365, 375, 377, 378, 402, 505, 544, 546, 547                    | 1.122 |
| 33 | 1jszA_BS01_SAH | 39,42,66,67,68,69,70,71,72,74,95,97,115,116,138,139,140,153                         | 39,42,66,69,70,71,72,79,95,97,115,116,138,139,140,153               | 1.447 |
| 34 | 1ju2B_BS01_UUU | 116, 131, 132, 135, 171, 176, 178                                                   | 116, 131, 132, 176, 178                                             | 3.6   |
| 35 | 1jx4A_BS05_ADI | 7,8,10,11,12,44,45,51,57,76,105,159                                                 | 7,8,10,11,12,44,45,51,57,76,105,159                                 | 0.776 |
| 36 | 1ka1A_BS03_A3P | 71, 141, 144, 145, 146, 239, 240, 262, 263, 266, 280, 287, 293                      | 71, 141, 144, 145, 146, 239, 240, 262, 263, 266, 280, 287, 293      | 2.319 |
| 37 | 1kgqA_BS01_NPI | 104,124,139,141                                                                     | 104,124,139,141                                                     | 1.557 |
| 38 | 1kgqA_BS02_SCO | 183, 185, 186, 187, 203, 204, 217, 234, 253, 254, 259, 260, 263                     | 183, 185, 217, 234, 253, 254, 259, 260, 263                         | 4.782 |
| 39 | 1kt8B_BS01_ILP | 75,77,99,141,143,192,202,207,237,240,241,242,266,268,269,270,312,313                | 75,77,99,192,202,207,237,240,241,242,266,268,269,270,312,313        | 1.311 |
| 40 | 1ktgA_BS04_AMP | 28,30,35,75,82,120                                                                  | 28,30,35,75,83,120                                                  | 1.523 |
| 41 | 1kwfA_BS01_BGC | 52,61,63,100,113,116,117,118,120,169,172,173,183,245,246,249,337,339,340            | ,116,117,118,120,169,172,249,337,339,340                            | 6.354 |
| 42 | 1lwdB_BS01_ICT | 212,252                                                                             | 212,252                                                             | 1.121 |
| 43 | 1mv8C_BS01_NAD | 7,9,10,11,29,30,31,35,83,84,85,86,98,102,105,123,124,157                            | 7,9,10,11,29,30,31,102,105,123,124,157                              | 0.905 |
| 44 | 1n1dB_BS02_C2G | 8,9,10,17,20,44,46,71,74,77,91,92,94,95,99,113,114,117                              | 8,9,11,17,20,44,46,74,75,77,91,92,94,95,99,113,114,118              | 3.198 |
| 45 | 1nfzB_BS03_EIP | 18,48,52,64,65,66,80,84,101,111,113,158                                             | 18,48,52,64,65,101,111,113,158                                      | 1.876 |
| 46 | 1o03A_BS01_G16 | 8,9,10,20,46,47,49,52,114,115,116,117,118,145                                       | 8,9,10,20,46,47,49,52,114,115,116,117,118,145                       | 1.588 |
| 47 | 1o94A_BS02_ADP | 395, 396, 398, 399, 400, 419, 420, 426, 427, 469, 470, 486, 487, 488, 673, 674      | 395, 396, 398,400, 420, 426, 427, 469, 470, 486, 487, 488, 673, 674 | 1.064 |
| 48 | 1ohaA_BS02_NLG | 43,44,45,65,66,80,122,147,158,159,160,161                                           | 43,44,45,65,66,80,122,147,158,159,160,161                           | 0.814 |
| 49 | 1pjaA_BS04_DXE | 244,269,287                                                                         | 244,265,283                                                         | 0.546 |
| 50 | 1pjaA_BS06_MXE | 213,214                                                                             | 212,214                                                             | 0.375 |



Figure 2.1 shows the histogram of RMSD values from the docking simulations of 50 protein-ligand complexes. RMSD is also an important index for evaluating the structural correctness of the docked predictions, which measures the docking poses relative to the standard binding conformations. As a rule of thumb, an RMSD  $<2.0$  Å is an acceptably low deviation between predicted and known experimental binding mode.



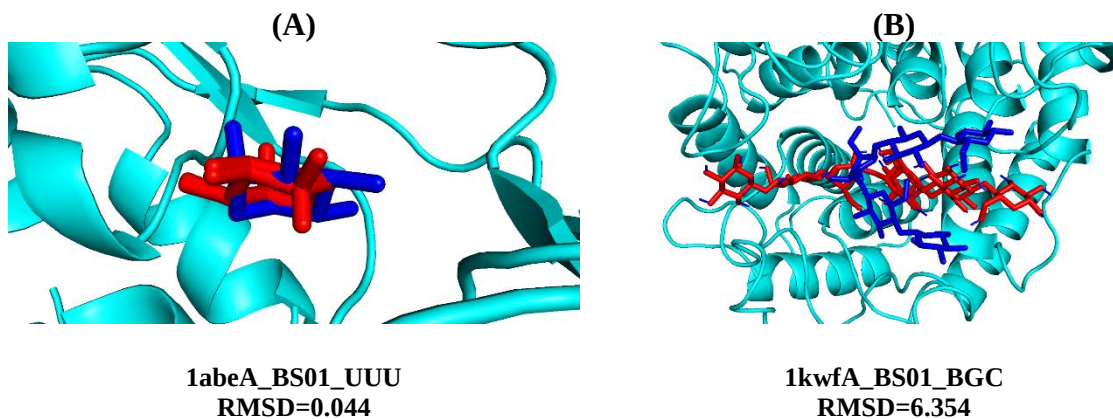
**Figure 2.1** A histogram of the RMSD values fitted to a skewed distribution. The x- and y-axes correspond to RMSD values and their frequency, respectively. The distribution of the RMSD values is shown as the blue curve. The red dashed vertical line represents a particular RMSD threshold.

As shown in Figure 2.1, the distribution of RMSD values appears moderately right-skewed, with a majority of the values concentrated in the 0-2 Å range, and fewer entries extending toward higher values. A red dashed line at 2.0 Å indicates the threshold used to

assess docking accuracy. Most complexes lie to the left of this threshold, which confirms that the docking results and the conformational prediction method constitute a reliable approach.

The kernel density estimate (KDE) overlaid on the histogram provides a smoothed representation of the distribution, confirming the inference of one peak in the low RMSD range (around 0.5-1.5 Å). Only a few complexes exceed 3 Å, indicating potential discrepancies in those specific dockings, possibly due to flexible ligands, unusual binding pockets, or limitations in scoring functions.

Overall, the distribution indicates that the docking protocol and binding site identification using AutoDock Vina are stable. The majority of RMSD values are within acceptable limits, confirming the reliability of binding site predictions. Visual and numerical evidence for the validity of the docking results is available from this analysis.



**Figure 2.1.** Structural alignment of ligand poses within binding sites from two protein-ligand complexes. (A) shows that the complex 1abeA\_BS01\_UUU has an RMSD of 0.044. (B) shows the complex 1kwfA\_BS01\_BGC with an RMSD of 6.354. Protein backbones are shown in cyan, ligands in red (reference) and blue (target).

Further statistical analysis was conducted to examine the relationship between RMSD values and several factors affecting docking accuracy, including ligand size, receptor flexibility, and binding affinity. The analysis revealed that larger and more flexible ligands, as shown in Figure 2.1, typically resulted in higher RMSD values (1kwfA\_BS01\_BGC RMSD = 6.354), suggesting that the docking algorithm struggled more with these types of ligands. Larger ligands tend to have more possible conformations, which complicates the docking process and leads to less accurate predictions. Conversely, smaller and more rigid ligands tended to produce lower RMSD values, as shown in Figure 2.1(1abeA\_BS01\_UUU RMSD=0.044), indicating that AutoDock Vina was more successful in docking these ligands accurately.

Moreover, receptor flexibility was another crucial factor influencing the accuracy of docking. Highly flexible receptors that could adopt multiple conformations resulted in higher RMSD values, as the docking algorithm struggled to predict accurate poses for these flexible receptors. On the other hand, rigid receptors (1abeA\_BS01\_UUU is more rigid than 1kwfA\_BS01\_BGC) with a more stable binding site led to more accurate docking predictions, resulting in lower RMSD values.

Given that receptor flexibility significantly contributed to increased RMSD values, it became evident that static docking alone may not fully capture the dynamic nature of protein-ligand interactions. The study proposes transitioning to molecular dynamics (MD) simulations as a complementary approach to address this limitation. Unlike rigid docking models, molecular dynamics (MD) allows for exploring receptor conformational changes over time, offering a more realistic depiction of the binding environment. By incorporating receptor motion and solvent effects, molecular dynamics (MD) simulations can provide deeper insights

into ligand binding stability and help refine docking results, particularly when high flexibility hinders accurate pose prediction. This shift toward dynamic modeling represents a crucial step forward in improving the reliability of computational drug discovery pipelines.

### 3.6 Conclusion

The analysis of RMSD distributions from docking simulations indicates that AutoDock Vina produces accurate and reliable predictions for most protein-ligand complexes, with the majority of RMSD values falling within the accepted 2.0 Å limit. This attests to the portability of the docking protocol and the efficiency of the approach used for the identification of the binding site. The variations observed in docking accuracy can mainly be due to factors such as ligand size and flexibility, and the conformational flexibility of the receptor. Larger and more flexible ligands and flexible receptors pose greater challenges for accurate docking, resulting in higher RMSD values, whereas smaller, rigid ligands and stable receptor sites facilitate more precise binding predictions. Overall, these findings validate the utility of AutoDock Vina as an effective molecular docking and virtual screening tool, which can generate valuable information on ligand-receptor interactions to assist in drug discovery searches. These results demonstrate that the docking parameters consistently and efficiently perform docking against a range of protein-ligand complex structures.

**Section B:**

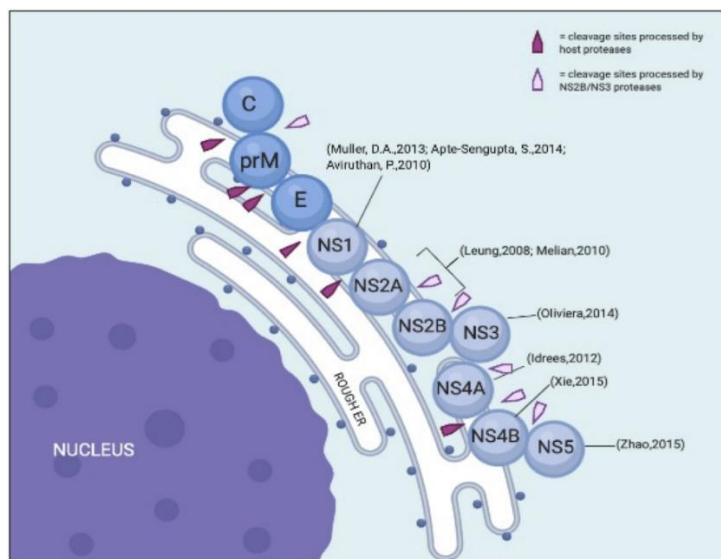
**Application of the Galaxy BRIDGE**

**Tool**

## CHAPTER 3

### 4.1 Dengue Virus

Dengue fever is one of the most significant public health concerns worldwide, due to its rapid spread, high infection rates, and potential to trigger severe, life-threatening symptoms. Dengue is caused by the dengue virus (DENV), an RNA virus that belongs to the *Flavivirus* genus within the *Flaviviridae* family. It is predominantly transmitted to humans through the bite of DENV-infected *Aedes* mosquitoes, primarily *Aedes aegypti*. Dengue affects millions of people worldwide every year and is caused by four distinct yet closely related serotypes (DENV-1 to DENV-4), which share approximately 65-70% of their genomic sequences. This genetic diversity allows for the possibility of multiple directly infecting strains in the same host, resulting in successive infections that lead to a tendency for more severe clinical manifestations. The most common serotype is DENV-1, with DENV-2 often associated with severe forms of disease. The more severe, life-threatening conditions known as dengue hemorrhagic fever (DHF) and dengue shock syndrome (DSS) can lead to internal bleeding, organ damage, and death if not treated. Although it is a global burden, there are no specific antiviral treatments for dengue; current clinical management is largely supportive, highlighting the critical need for efficacious interventions. There is no universally reliable vaccine to have, and the disease's public health impact is worsening, thanks to factors such as urbanization, climate change, and growth in global travel.<sup>24</sup>



**Figure 3.1** Illustration of DENV structural and NS proteins on the ER with their cleavage sites<sup>16</sup>

Morphologically, dengue virus measures 50 nm in diameter and contains a positive-sense single-stranded RNA genome of 10 kb or more, which encodes a single open reading frame (ORF). Upon infection, the viral RNA is replicated and translated in the endoplasmic reticulum, where the host ribosomes translate a single polyprotein into a protein. This polyprotein is cleaved co- and post-translationally by host and viral proteases into three structural proteins (capsid (C), envelope (E), and membrane (M)) and seven nonstructural (NS) proteins (NS1, NS2A, NS2B, NS3, NS4A, NS4B, and NS5). The structural proteins comprise the virus particle, and the NS proteins are essential for viral replication, protein processing, alteration of host cell function, and immune evasion. Of the NS proteins, NS3, with its protease and helicase activities, and NS5, as an RNA-dependent RNA polymerase and methyltransferase, are essential for viral replication and hence represent the most promising candidates for antiviral drug development.<sup>16,25</sup>



In particular, the NS2B-NS3 protease is responsible for cleaving the viral polyprotein and facilitating the virus's maturation. Its crystal structure (PDB ID: 2FOM) was thoroughly studied. It serves as a starting point for computer-aided drug design, such as molecular docking and molecular dynamics simulations, to screen for inhibitors that interfere with viral replication. Although several candidate drugs targeting various stages of the dengue viral life cycle, such as protease inhibitors (aprotinin and peptidomimetics) and polymerase inhibitors (sofosbuvir, a drug repurposed from hepatitis C treatment), have been developed, none have successfully passed the stage of full regulatory approval. Entry and fusion inhibitors, including ligands interfering with viral binding and membrane fusion, as well as host-directed approaches targeting cellular pathways critical for viral propagation, are also being actively explored. A number of compounds, such as celgosivir, balapiravir, and favipiravir, advanced to clinical trials, but none proved to be efficacious in the largest studies. Nevertheless, recent progress in structure-based drug design and computational screening is still enabling the identification and optimisation of targeted antiviral agents, which promise the development of effective dengue therapeutics.<sup>26</sup>

## **4.2 Natural Products**

Plants and various other organisms produce a huge and diverse array of chemical compounds collectively known as natural products or secondary metabolites. While primary metabolites are essential for basic cellular function and growth, secondary metabolites perform specialized ecological functions that help an organism to interact with its environment. Once considered mere metabolic waste, these compounds gained recognition with the rise of chemical ecology in the late 20th century, revealing their critical roles in defense against herbivores and pathogens, interspecies communication, and attraction of pollinators through

pigmentation and scent. Some natural products function as signaling molecules and plant hormones that mediate intricate physiological responses within the organism. In addition to ecological functions, natural products have played a profound role in human culture and health since ancient times, serving as traditional medicines, flavorings, and dyes. With the advancement of modern pharmacology, their significance has intensified, as natural products isolated from terrestrial plants, marine organisms, and microbes continue to be foundational in the discovery of new drugs. Due to their extremely high structural diversity and unusual scaffolds, which are seldom found in synthetic compound libraries, a substantial portion of FDA-approved drugs have been derived directly or indirectly from natural sources.<sup>27</sup>

Natural products typically have complex structures assembled through biosynthetic pathways that involve enzymes and precursor molecules, such as acetyl-CoA, amino acids, and isoprenoids. They are broadly categorized into classes, including alkaloids, terpenoids, flavonoids, polyketides, steroids, and glycosides. The biosynthetic complexity of these compounds presents an enormous opportunity for medicinal chemistry, particularly in combination with modern approaches such as genome mining, dereplication, and synthetic biology.<sup>28</sup> Today, natural product research is a vibrant and multidisciplinary domain that encompasses ethnopharmacology, analytical chemistry, molecular biology, and computer science. As the exploration of the chemical diversity of nature proceeds, natural products continue to offer an indispensable source of drug leads, particularly in the area of combating infectious diseases, such as dengue, and other urgent global health threats.<sup>28</sup>

For the discovery and development of bioactive natural compounds, dedicated natural product databases have been established as indispensable bioinformatics resources. These databases contain structured data on chemical structures, biological activities, pharmacological

activities, and the producing organism, facilitating effective virtual screening, molecular docking, and in silico drug discovery. Well-known databases include NPASS, which links compound bioactivity with the original species; SuperNatural II, a large set of diverse natural compounds; the ZINC Natural Products subset, featuring purchasable natural-like molecules; AfroDB, with an emphasis on African medicinal plants; and IMPPAT, which lists phytochemicals from Indian medicinal resources. These databases, which contain curated, biologically relevant compound libraries and corresponding pharmacological or literature information, can effectively expedite and broaden the profile of early drug discovery in structure-based drug design and virtual screening.

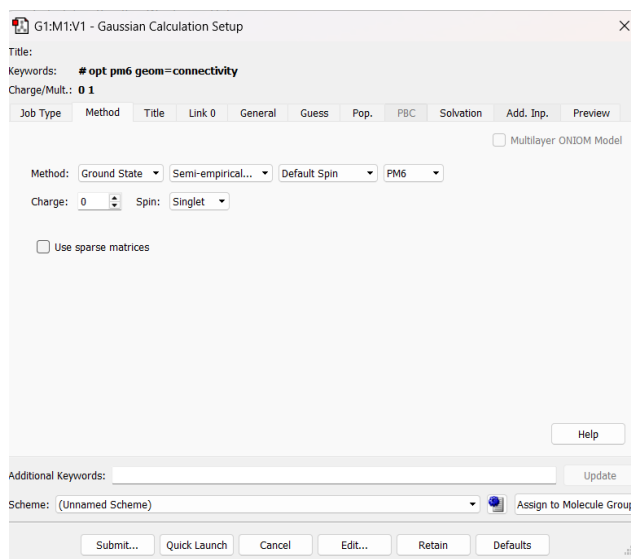
### **4.3 Natural Product Database**

Lead optimization in the search for potential dengue-virus inhibitors required several systematic steps, including ligand selection from diverse biological sources, refining their structures, and preparing these refined molecules for molecular docking studies. This step has also been involved in the preparation of the chosen molecules obtained from the PubChem database and other well-known databases for assessing the selected drug candidates for the dengue virus protein targets.

The first step is followed by the identification of a diverse set of biologically relevant, chemically feasible molecules suitable for antiviral studies. Based on their general bioactivity profiles, 572 compounds were selected, all of which exhibited potential antiviral, anti-inflammatory, antioxidant, or related bioactivities. The compounds were produced by various plant species and marine organisms, each possessing unique mechanisms of action, such as reducing viral load, inhibiting viral entry, inhibiting oxidative stress pathways, or modulating

the immune system. The molecules were also selected from well-established chemical databases, including PubChem<sup>29</sup> and additional sources that document the compounds' bioactivity and mechanism of action. These molecules were arranged in a table for easy reference, containing important details including compound name, source species, bioactivity, mechanism of action, target protein/pathway, availability in Sri Lanka, and recorded chemical structures.

For instance, molecules such as 6-Gingerol from *Zingiber officinale* have been reported to inhibit viral load by targeting NS3 helicase. Similarly, molecules such as 7-Epiculisinone from *Rheedia brasiliensis* were suggested to inhibit viral entry. Additional compounds (e.g. 7-Hydroxyflavone) were extracted from different kinds of plants rich in flavonoids [32] because of their capacity to block oxidative stress pathways. All of these molecules were evaluated based on their bioactivity and availability, primarily in Sri Lanka, to determine if they were suitable for the next stage of experimental verification. The specific structures and bioactivity data were critical to the interpretation of how these compounds could possibly interact with the dengue virus.



**Figure 3.2** Gaussian Calculation Setup used to create Natural Product Database.

After selecting the compounds, the 3D molecular structures of these compounds were optimized for accurate docking studies in their stable conformation. This was done using the Gaussian software<sup>30, 31</sup>, which integrates efficient computational chemistry to optimize the geometry of each molecule. Optimization was carried out using the semi-empirical PM6 method to strike a balance between computational cost and accuracy in representing very large molecular systems.

An average potential energy calculation requires several extremely costly evaluations of two-electron integrals. Alternatively, various semiempirical methods are formulated to apply the assumptions and empirical parameters, thereby reducing the number of two-electron integrals. The most frequently used assumption, the zero differential overlap assumption, discards all products of the basis functions that are located on other atoms. This assumption allows the overlap matrix in the secular equation to be reduced to the unit matrix, one-electron integrals involving three centers are set to zero, and all three and four-center electron integrals

can be neglected. Therefore, methods that use the zero differential overlap assumption are known as neglected integral methods. Standard semiempirical methods include the Complete Neglect of Differential Overlap (CNDO), Intermediate Neglect of Differential Overlap (INDO), Modified INDO (MINDO), Austin Model 1 (AM1), and Parameterized Model 3 (PM3) and 6 (PM6).<sup>32</sup>

The PM6 (Parameterized Model number 6) method belongs to the family of NDDO (Neglect of Diatomic Differential Overlap) techniques, which approximate molecular electronic structure calculations by simplifying integrals and using empirical parameters such as derived from experimental data and high-level ab initio calculations. PM6 is an upgrade of previous semi-empirical studies (PM3, AM1), providing improved accuracy over a wider variety of molecular systems, ranging from organic molecules to transition metals and biomolecules, through re-parameterization of the integrals and a larger fit database. In PM6, the electronic Hamiltonian is simplified by neglecting certain two-electron integrals involving different atomic centers, reducing computational complexity. The total electronic energy  $E$  of a molecule is approximated as

:

$$E = \sum_i \langle \psi_i | \hat{H} | \psi_i \rangle + \sum_{A < B} \frac{Z_A Z_B}{R_{AB}} + \text{parameter corrections} \quad (3.1)$$

Where  $\psi_i$  are molecular orbitals,  $\hat{H}$  is the Hamiltonian operator,  $Z_A$  and  $Z_B$  are nuclear charges of atoms A and B, and  $R_{AB}$  is the distance between nuclei A and B. Parameter corrections account for empirical adjustments that improve agreement with experimental results. The PM6 method parameterizes core-core repulsion functions and adjusts resonance

integrals to reproduce heats of formation, geometries, ionization potentials, and other molecular properties across a wide range of training sets.<sup>33</sup>

The primary advantage of PM6 is its balance between computational efficiency and reasonable accuracy, making it suitable for optimizing large molecules and performing geometry optimizations quickly compared to more computationally demanding *ab initio* or density functional theory (DFT) methods. PM6 is especially useful in early-stage drug discovery and large-scale screening, where thousands of compounds require rapid evaluation. Its improved parameterization over previous methods results in better handling of hydrogen bonding, non-covalent interactions, and transition metal complexes, which are critical for biological and organometallic systems.

However, PM6 is semi-empirical and thus has its constraints. The method relies heavily on empirical parameters fitted to a specific set of molecules, it cannot be used to make precise predictions for systems that are quite different from the training data, such as those with unusual bonding environments or exotic elements. PM6 usually yields poorer energetics and electronic properties compared to high-level *ab initio* or density functional theory (DFT). It may also struggle to describe the dispersion interaction and the subtle electronic effects that are important in certain electronic properties of biomolecular systems. Despite these limitations, PM6 still represents a useful compromise in computational chemistry where fast geometry optimizations and electronic structure approximation of medium- to large-size molecular systems are required in a computationally feasible manner. The optimization process was crucial, as only molecules in their lowest-energy state are expected to exhibit stable binding interactions during docking with the dengue virus target protein.

The optimization of individual ligand structures involved the next step, which was compiling all the optimized ligands into a single, unified file for use in molecular docking simulations. Schrodinger software<sup>34</sup> was employed for this task due to its robustness in handling large sets of molecules. The optimized SDF files of each ligand were imported into Schrodinger and then organized into a single SDF file. This compilation was crucial in reducing the complexity of the docking process and facilitating the simultaneous screening of 572 ligands in subsequent virtual screening and docking studies.

The SDF format of the molecules (with coordinates in 3D space and chemical properties) was selected because most molecular docking computer programs can easily read it. Schrodinger also consolidated the ligands into a single file so no data was lost, and batch processing for docking studies became more straightforward. Having a single, unified way to generate this information meant that there was no longer the hassle of dealing with multiple files during docking simulations, making it easier to screen all compounds more rapidly.

The final product was a single SDF file with 572 optimized ligands. This file was used in a molecular docking study to analyze the binding affinity of ligands with dengue virus protein targets. The molecular docking studies, in which the selected ligands, acting as non-nucleoside inhibitors, were docked into the docking pocket of the NS3 protease or helicase, or into other antiviral pathways of the viral proteins. The objective was to identify compounds that could bind to the virus and inhibit its replication or entry, thereby acting as potential lead compounds for experimental testing. The ligands were sorted according to their docking scores in descending order (the larger the score, the greater the binding strength). Compounds ranking above the threshold were considered for additional evaluation.



#### 4.4 Computational Methods

The binding affinities and probable efficacies of selected ligand molecules against dengue virus targets were calculated using computational methods presented in this thesis, with a focus on the NS2B-NS3 protease. The initial binding poses of ligands in the active site of the protein were predicted based on RMSD values obtained after molecular docking, and these values were calculated using Galaxy BRIDGE tools. Enhanced sampling and more precise computation of binding free energies were performed by employing MM-GBSA and MM-PBSA. These approaches offer a more accurate thermodynamic analysis, allowing for the inclusion of solvation contributions and the energetic costs for proteins and ligands in both complex and uncomplexed forms. MM-GBSA and MM-PBSA were applied to the top docking poses to rank ligand-protein complexes based on their estimated binding free energies ( $\Delta G_{\text{binding}}$ ). Additionally, hydrogen bond analysis was also carried out to identify the principal stabilizing contributors to the interaction between ligand and active site residues, providing information about the specificity and strength of binding interactions. This analysis facilitated the identification of ligands that establish a strong and directional hydrogen bonding network, which typically plays a substantial role in binding affinity and biological activity. In silico toxicity prediction, aside from binding energy analysis, was performed with ProTox3. This involved predicting their parameters, such as hepatotoxicity, carcinogenicity, mutagenicity, and LD<sub>50</sub> values. As a result, the compounds not only exhibit strong vein-binding affinities but also possess reliable safety profiles. Computational approaches like these were used to identify candidates for in vitro and in vivo validation as dengue virus inhibitors.

#### 4.5 Molecular Dynamics Simulation.

Molecular Dynamics (MD) simulation is a powerful method for modelling atomic and molecular motions by solving Newton's equations of motion electronically. It is broadly used in various fields, including structural biology and drug design, to investigate the dynamic properties of molecular systems. The basic equation underlying MD is Newton's second law, expressed as:

$$F_i = m_i a_i = m_i \frac{d^2 r_i}{dt^2} \quad (3.2)$$

Where  $F_i$  is the force on atom  $i$ ,  $m_i$  is its mass,  $a_i$  is its acceleration, and  $r_i$  is its position vector. These forces are derived from a molecular mechanics force field that describes the potential energy  $U$  of the system<sup>35</sup>. The potential energy function includes both bonded and non-bonded interactions and can be expressed as:

$$E_{total} = E_{bonded} + E_{non-bonded} \quad (3.3)$$

$$E_{total} = \sum_{bonds} k_b (r - r_0)^2 + \sum_{angles} k_\theta (\theta - \theta_0)^2 \\ + \sum_{dihedrals} V_n [1 + \cos(n\phi - \gamma)] + \sum_{i < j} \frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^6} + \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} \quad (3.4)$$

The first three terms were those of bonded interaction i.e., bond stretching, angle bending and torsional rotations and last terms were non-bonded van der Waals (Lennard-

Jones) and electrostatic interactions. The positions and velocities of the atoms are iteratively updated using numerical integration methods, such as the Verlet or velocity-Verlet method. For instance, the Verlet integration is a scheme in which the update of the positions is described as:

$$r_i(t + \Delta t) = 2r_i(t) - r_i(t - \Delta t) + \frac{F_i(t)}{m_i} \Delta t^2 \quad (3.5)$$

To maintain temperature and pressure in biological systems, MD employs thermostats and barostats. A popular thermostat is the Nosé-Hoover thermostat, which introduces a heat bath by modifying the equations of motion using an extended Lagrangian approach.<sup>36</sup> The kinetic energy  $K$  and potential energy  $U$  of the system together form the Hamiltonian  $H$ , which represents the total energy:

$$H = K + U = \sum_i \frac{1}{2} m_i v_i^2 + U(r_1, r_2, \dots, r_N) \quad (3.6)$$

Here,  $v_i$  is the velocity of atom  $i$ , and  $U$  is the potential energy as a function of all atom positions. MD simulations produce time series of atomic positions (trajectories), and they can be examined for various properties. Popular analysis measures are the Root Mean Square Deviation (RMSD) and the Root Mean Square Fluctuation (RMSF):

$$RMSF_i = \sqrt{\langle (r_i(t) - \langle r_i \rangle)^2 \rangle} \quad (3.7)$$

These measure the structural deviation and flexibility of molecules during the simulation. MD has become indispensable in drug discovery by allowing researchers to observe ligand-protein interactions in dynamic detail. It can aid better, less ambiguous binding free energy calculations using the approaches MM/PBSA and MM/GBSA.

#### **4.6 Explanation Of Binding Free Energy Calculation Using Molecular Mechanics and Solvation Models, Including Polar and Nonpolar Contributions.**

Binding free energy ( $\Delta G_{\text{bind}}$ ) is an essential parameter in computer-aided drug discovery, and it refers to the strength of interaction between a ligand (drug molecule, for instance) and a target protein. This free energy of binding is obtained by subtracting the free energy of the bound complex from the sum of the free energies of the unbound protein and ligand:

$$\Delta G_{\text{bind}} = G_{\text{complex}} - (G_{\text{receptor}} + G_{\text{ligand}}) \quad (3.8)$$

Equation 3.8 measures the thermodynamic favorability of the ligand binding to the protein; thus, more negative values for  $\Delta G_{\text{bind}}$  imply a tighter binding, which is often desired for drug design. For the complex, the protein, and the ligand, each free energy  $G$  can be decomposed into three basic terms: the molecular mechanics energy  $E_{\text{MM}}$ , the solvation free energy  $G_{\text{solvation}}$ , and the entropy term  $TS$  as:

$$G = E_{\text{MM}} + G_{\text{solvation}} - TS \quad (3.9)$$

The molecular mechanics energy (EMM) includes contributions due to bond stretching, angle bending, and torsional rotation, whereas nonbonded interactions describe van der Waals forces and electrostatic (Coulomb) interactions between atoms. The solvation free energy,  $G_{\text{solvation}}$ , is a measure of the energetic contribution of the surrounding solvent (typically water) to the molecule and how solvation either favors or opposes the structure. This solvation energy is decomposed into polar and nonpolar solvation energies as:

$$G_{\text{solvation}} = G_{\text{polar}} + G_{\text{nonpolar}} \quad (3.10)$$

Polar solvation energy accounts for the electrostatic interactions between the solute (protein or ligand) and the solvent, while the nonpolar solvation energy reflects hydrophobic effects and the energetic penalty of creating a cavity in the solvent to accommodate the molecule. The nonpolar term is usually empirically estimated as a function of the solvent-accessible surface area (SASA) of the molecule using the equation;

$$G_{\text{nonpolar}} = \gamma \cdot \text{SASA} + b \quad (3.11)$$

where  $\gamma$  is a surface tension coefficient (commonly around 0.0072 kcal/mol/Å<sup>2</sup>), and  $b$  is a small constant fitted to experimental data.

The entropy term TS (Equation 3.9), which accounts for changes in molecular flexibility and disorder upon binding, is often omitted or approximated due to the high computational cost of accurate entropy calculations.<sup>37</sup>

MM/GBSA and MM/PBSA are two widely used end-point methods for predicting binding free energies. Both are based on the decomposition of the overall active free energy into molecular mechanics and solvation terms but employ different methodologies for computing the polar solvation free energy.

MM/GBSA estimates the polar solvation energy using the Generalized Born (GB) model. The GB model is a fast, approximate, analytic approach for estimating the manner in which the solvent electrostatically screens interactions among charged atoms in the molecule. The polar solvation MM/GBSA is determined using:

$$G_{polar}^{GB} = -\frac{1}{2} \left(1 - \frac{1}{\epsilon}\right) \sum_{i,j} \frac{q_i q_j}{f_{GB}(R_{ij}, \alpha_i, \alpha_j)} \quad (3.12)$$

where  $q_i$  and  $q_j$  are the partial atomic charges,  $r_{ij}$  is the distance between atoms  $i$  and  $j$ ,  $\alpha_i$  and  $\alpha_j$  are the Born radii describing how buried each atom is within the molecule, and  $f_{GB}$  is an effective distance function devised to model how the strength of the electrostatic screening varies with the distance between atoms and with the level of solvation. While crude, this model is computationally inexpensive and can be applied to analyze large data sets or screen a large number of ligands.

In contrast to that, the polar solvation energy in MMPBSA has been computed using the Poisson-Boltzmann (PB) equation. The PB equation is a more physical model that solves numerically the electrostatic potential  $\phi(r)$  around the molecule, taking into account explicitly the space-dependent dielectric properties and the ionic strength of the solvent. The polar solvation energy in MMPBSA is expressed as:

$$G_{polar}^{PB} = \frac{1}{2} \int \rho(r) \phi(r) dr \quad (3.13)$$

where  $\rho(r)$  is the charge distribution of the density, while solving the PB equation yields more accurate and physically motivated electrostatic energies, it is computationally more demanding and far more time-consuming and intensive than the GB model.

The nonpolar solvation energy is treated similarly by MM/GBSA and MM/PBSA, namely through an empirical model based on the solvent-accessible surface area (SASA) as described above. This takes into account hydrophobic interactions and the free energy associated with the creation of a cavity in the solvent.

MM/GBSA and MM/PBSA are widely used end-point methods that can estimate the binding free energy, where the total energy is decomposed into molecular mechanics and solvation contribution. MM/GBSA allows for faster, though less accurate, calculations that employ the Generalized Born model for polar solvation, making it suitable for throughput screening and large-scale studies. MM/PBSA offers a more accurate (but computationally time-consuming) approach by solving the Poisson-Boltzmann equation, and is best suited for

detailed investigations that require high precision. The nonpolar solvation term is similarly treated by the solvent-accessible surface area models in both methods.

From a more fundamental perspective, combining these methods with current machine learning approaches may even enhance binding affinity predictions. For instance, one may train surrogate models to predict PB solvation energies using features engineered from GB calculations, with which accuracy rivalling MMPBSA and speed rivalling MM/GBSA are expected to be achieved. Furthermore, techniques used for active learning could focus in a cost-effective manner only on the most critical conformations with high PB calculations, enhancing efficiency. Hybrid strategies like them have the potential to expand the frontier of computational drug discovery.<sup>35</sup>

#### **4.7 Results and Discussion**

To evaluate the effectiveness and stability of the top ligand-protein complexes, several key computational analyses were performed. MM/GBSA and MM/PBSA binding free energy calculations were carried out to estimate the strength of interaction between each ligand and the target protein. These methods are widely used in computational drug discovery because they provide a more accurate assessment of binding affinity by incorporating both molecular mechanics and solvation effects. Performing these calculations helps identify which ligands are likely to form energetically favorable and stable complexes, which is essential for selecting promising candidates for further development.

RMSD (Root Mean Square Deviation) analysis was conducted to examine the structural stability of both the protein backbone and the ligand during molecular dynamics simulations. The backbone RMSD gives insight into whether the protein maintains its

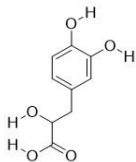
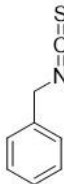
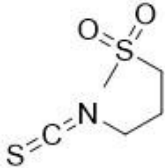
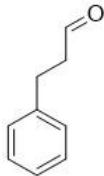
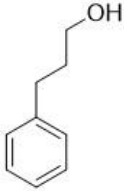
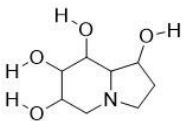


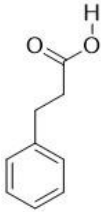
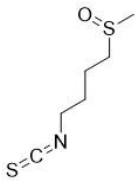
structural integrity over time, which is important for ensuring that ligand binding does not disrupt the overall protein conformation. Meanwhile, ligand RMSD helps determine how stably a ligand remains in the binding site throughout the simulation. If a ligand exhibits minimal movement, it suggests a strong and consistent interaction, which is a desirable property for a potential drug molecule.

Finally, toxicity prediction was included to assess the safety and drug-likeness of each compound. Computational toxicity screening helps identify any potential harmful effects, such as mutagenicity or organ toxicity, early in the drug discovery process. This step is essential to eliminate compounds that may show good binding but are unsuitable due to safety concerns.

Together, these analyses provide a thorough evaluation of the selected ligands in terms of their binding strength, structural stability, interaction profile, and predicted safety, supporting informed decision-making in the selection of potential drug candidates.

**Table 3.1** Docking scores and binding free energy calculations (MM/GBSA and MM/PBSA) for the ligand candidates targeting the dengue virus NS2B-NS3 protease.

| No | PubChem Id | Pose | Structure                                                                           | Docking Score (kcal/mol) | MM/GBSA     | MM/PBSA     |
|----|------------|------|-------------------------------------------------------------------------------------|--------------------------|-------------|-------------|
| 1  | 11600642   | 2    |    | -4.81                    | -12.41±1.93 | -6.61±2.12  |
| 2  | 2346       | 6    |    | -4.57                    | -18.49±0.48 | -13.57±0.40 |
| 3  | 3080557    | 2    |   | -4.51                    | -21.43±1.70 | -15.09±1.63 |
| 4  | 637511     | 8    |  | -4.45                    | -18.66±0.26 | -11.57±0.27 |
| 5  | 5315892    | 2    |  | -4.44                    | -20.31±1.50 | -13.70±1.47 |
| 6  | 5445       | 2    |  | -4.38                    | -0.69±5.25  | -0.30±5.09  |

|   |        |   |                                                                                   |       |             |             |
|---|--------|---|-----------------------------------------------------------------------------------|-------|-------------|-------------|
| 7 | 444539 | 5 |  | -4.31 | -21.09±1.84 | -14.65±1.58 |
| 8 | 5350   | 1 |  | -4.30 | -19.02±0.30 | -13.47±0.88 |

#### 4.8 Analysis of binding free energy calculations (MM/GBSA, MM/PBSA)

In this study, post-docking binding free energy refinement was performed in this study using MM/GBSA (Molecular Mechanics/Generalized Born Surface Area) and MM/PBSA (Molecular Mechanics/Poisson-Boltzmann Surface Area) approaches to better evaluate the binding affinity of the top 10 ligand candidates against dengue virus protease (Table 3.1). These approaches extend beyond rigid docking scores, incorporating molecular mechanics and implicit solvent models to provide more realistic and dynamic assessments of ligand-protein interactions. The MM/GBSA binding energies ranged from  $-0.69 \pm 5.25$  kcal/mol (ligand 5445) to  $-21.43 \pm 1.70$  kcal/mol (ligand 3080557), showing substantial variation in predicted binding strength. Notably, ligands 3080557, 444539, and 5315892 consistently ranked among the strongest binders across both MM/GBSA and MM/PBSA evaluations despite moderate docking scores, suggesting that these molecules may achieve greater stability and favorable energetic profiles during molecular dynamics simulations than indicated initially by docking.

On the other hand, ligand 5445, which, except for docking score ( $-4.38$  kcal/mol) showed both the MM/GBSA ( $-0.69 \pm 5.25$  kcal/mol) and MM/PBSA ( $-0.30 \pm 5.09$  kcal/mol)

energies extremely weak, also indicating a lack of binding stability or unfavorable solvation. This highlights a key limitation of docking alone, where favorable poses can sometimes mislead without accounting for complete energetic realism. The wide standard deviations in ligands such as 11600642 (MM/GBSA:  $\pm 1.93$ , MM/PBSA:  $\pm 2.12$ ) and 5445 also suggest flexible or unstable binding modes, which would thereby cast doubt on these compounds as reliable inhibitors. In contrast, ligands such as 2346 and 637511 exhibited relatively low deviations and consistently favorable binding energies across both methods, reinforcing their potential as stable and energetically compatible candidates.

This comparative study also shows that MM/PBSA results are consistent with MM/GBSA trends while allowing the former to obtain slightly less favorable binding energies due to the different treatment of the solvation-free energy (Poisson-Boltzmann vs. Generalized-Born). Thus, the inclusion of MM/GBSA and MM/PBSA rescoring allowed us to further select more reliable inhibitors by eliminating false positives and enhancing the thermodynamic profile of ligands. Therefore, compounds 3080557, 5315892, and 444539 are top priorities due to their good binding affinities and dynamical stability, which may serve as a lead in subsequent optimization or experimental tests. This emphasizes the need for (structure-based) drug design pipelines that include more sophisticated free energy methods for a systematic and biologically more relevant compound prioritization.

## 4.9 RMSD Calculation

Root Mean Square Deviation (RMSD) is a quantitative measure of the average distance between atoms (typically heavy atoms) of a molecule over time, compared to a reference structure, usually the initial frame of a molecular dynamics (MD) simulation. In MD simulations, RMSD is a crucial parameter for evaluating the structural stability and motion of both the protein and the ligand within the simulation. RMSD computation offers information regarding the stability and mutual flexibility of the complex compared to its corresponding native conformation.

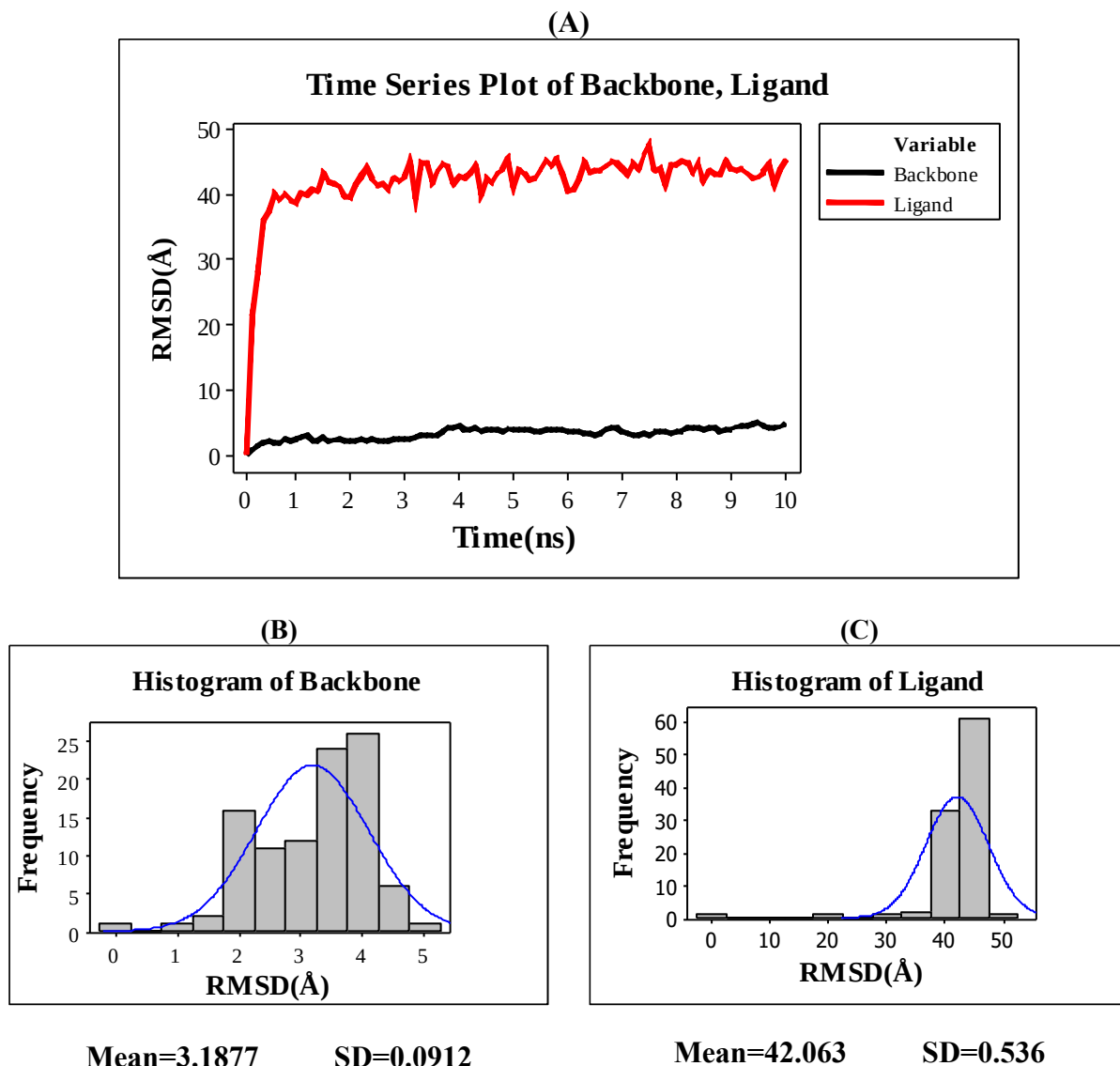
The top 10 ligand-protein complexes were selected in this study for the calculation of backbone RMSD and ligand RMSD. The main chain atoms (usually N, C $\alpha$ , and C) of the protein are emphasized in the backbone RMSD. This is evidence of the conformational stability of the protein upon binding the different ligands during the simulation. If, throughout the entire trajectory, the backbone RMSD remains quite low and does not change significantly (e.g., fluctuating between approximately 1-3 Å), the protein is considered to be in its native structure and remains stable across the simulation time. Otherwise, if the RMSD fluctuates significantly, it could imply structural instability or significant conformational change of the protein, e.g., the ligand binding.

The ligand RMSD, on the other hand, tracks the movement of the ligand molecule within the binding site over time. This is crucial because a ligand that remains tightly bound and exhibits minimal RMSD variation is likely to be a strong and stable binder. For such a ligand with a high fluctuating RMSD, if applicable, this could mean that the ligand doesn't quite fit into the binding pocket and is either being displaced or making non-native contacts.

This is likely due to the ligand having low binding affinity or incorrect geometry for effective binding.

Analysis of both backbone and ligand RMSD can provide a broader comprehension of each complex. A good drug candidate should be characterized by low values of both ligand RMSD (a measure of ligand binding stability) and backbone RMSD (a measure of protein structural stability). Thus, this dual analysis helps position and validate the performance of the top 10 ligands in both binding to the protein target and influencing the protein's dynamics. This is particularly relevant to drug discovery, as it helps to discriminate ligands well bound and/or consistently bound, these being the most likely to act as effective inhibitors in biological systems.

#### 4.9.1 Salianic acid (11600642)

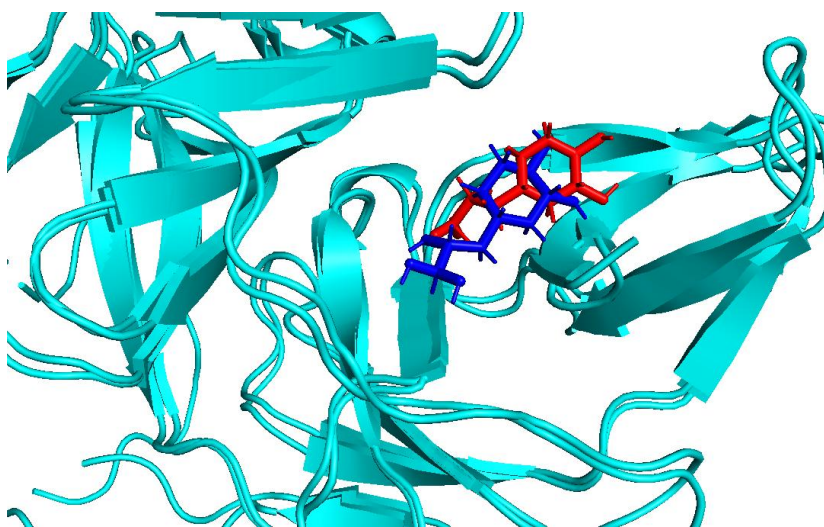


**Figure 3.3** RMSD analysis of the 10 ns molecular dynamics simulation with salianic acid (A). The time series plot of RMSD of the protein backbone (black) and ligand (red). Histograms display the distribution of RMSD values for the backbone (B) and ligand (C).

The RMSD (Root Mean Square Deviation) analysis, conducted over a 10-ns molecular dynamics simulation, provides important insights into the stability and dynamic behaviour of the protein-ligand complex. In the time series plot (Figure 3.3 (A)), the RMSD values of the protein backbone (black) and ligand (red) are clearly visible over the simulation time. The

protein backbone RMSD value remains at a low level, fluctuating mildly around the average value of  $\sim 3.19$  Å, indicating that the protein maintains a rigid and stable conformation throughout the simulation. On the other hand, the ligand RMSD starts to grow immediately in the first nanoseconds and becomes significantly higher, fluctuating at approximately 40 Å for the remainder of the simulation. This indicates a significant conformational adjustment and increased mobility of the ligand in the binding site as it explores various binding modes and orientations.

Histograms of the RMSD value distributions are shown in Figure 3.3 (B) and (C), and the backbone RMSD distribution (B) is relatively tight, peaking at around 3.2 Å with a small standard deviation of  $\sim 0.09$  Å, indicating a high structural rigidity of the protein. The ligand RMSD histogram, Figure 3.3 (C) exhibits a wider distribution, with an average distance of 42 Å and a standard deviation of 0.54 Å, indicating greater flexibility and dynamics of the ligand within the binding site during the simulation.

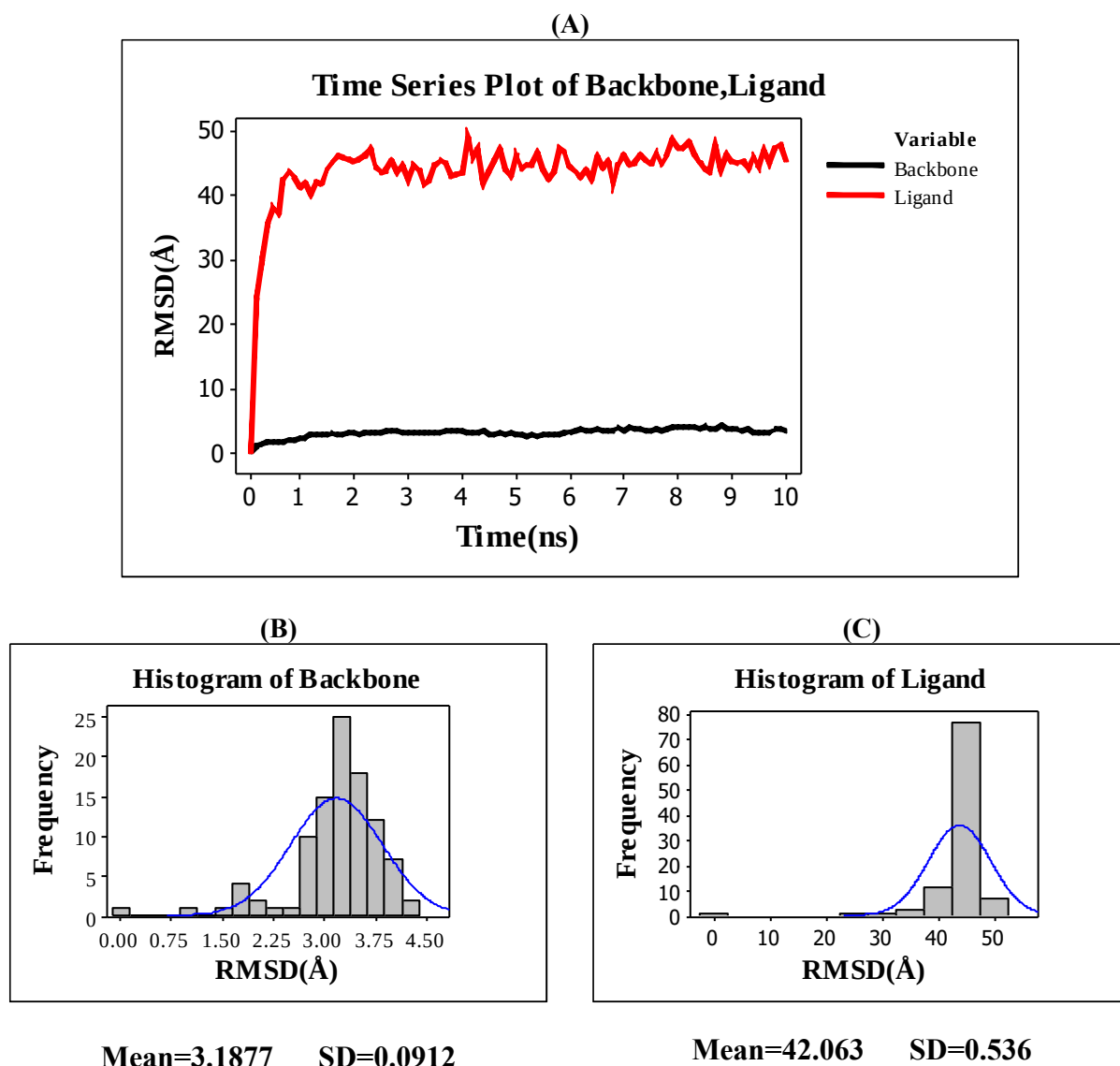


**Figure 3.4** The initial conformation of salianic acid at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket.



The RMSD data, presenting the ligand's dynamic behavior, are illustrated in Fig. 3.4. The ligand conformation at the first nanosecond is shown in red, representing the docked pose. On the other hand, the blue ligand conformation at the final nanosecond (10 ns) indicates a new, improved position resulting from the conformational adaptations in the simulation, which is energetically more favorable. The cyan-shaded protein scaffold remains virtually unaltered, validating the overall stability of the backbone suggested by the RMSD plot. This induced fit movement of the ligand is essential for generating optimal protein-ligand intermolecular interactions, representing the ligand's ability to accommodate itself inside the binding pocket in the most favorable position to improve affinity. These results reveal the rigidity of the protein structure and the flexibility and dynamism of the ligand, which are highly significant factors for understanding the binding mechanism and potential binding efficacy of the ligand in biological systems.

#### 4.9.2 Benzylisothiocyanate (2346)



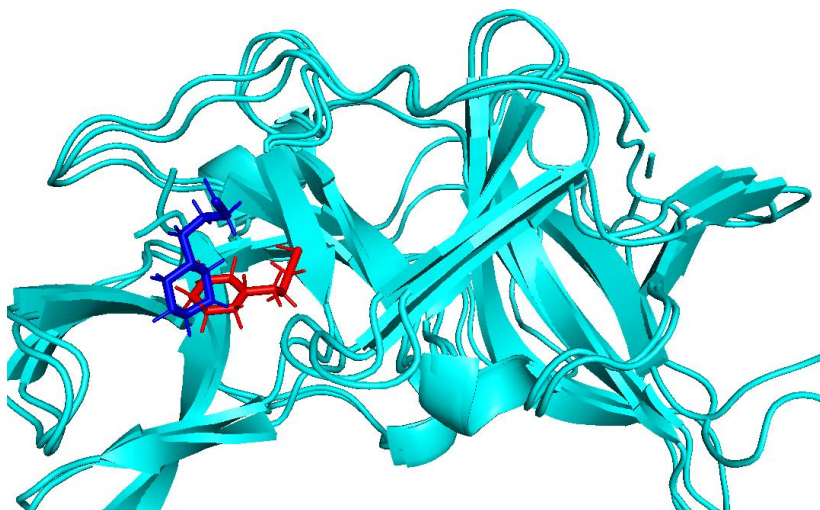
**Figure 3.5** RMSD analysis of the 10 ns molecular dynamics simulation with Benzylisothiocyanate (A). The time series plot of RMSD of the protein backbone (black) and ligand (red). Histograms display the distribution of RMSD values for the backbone (B) and ligand (C).

Based on Figure 3.5 (A), the RMSD of the protein's backbone remains at a low and relatively invariant level, with an average value of approximately 3.19 Å and a standard

deviation of 0.09 Å. This indicates that the protein structure is stable and consistent throughout the simulation process.

By comparison, the ligand quickly increases the RMSD and stabilizes at an average of approximately 42.06 Å, with a standard deviation of 0.54 Å, for the remainder of the simulation. This large increase and variability in the ligand RMSD indicate a significant conformational rearrangement and mobility in the binding site, as the ligand leaves its initially docked position and explores multiple energetically favorable conformations.

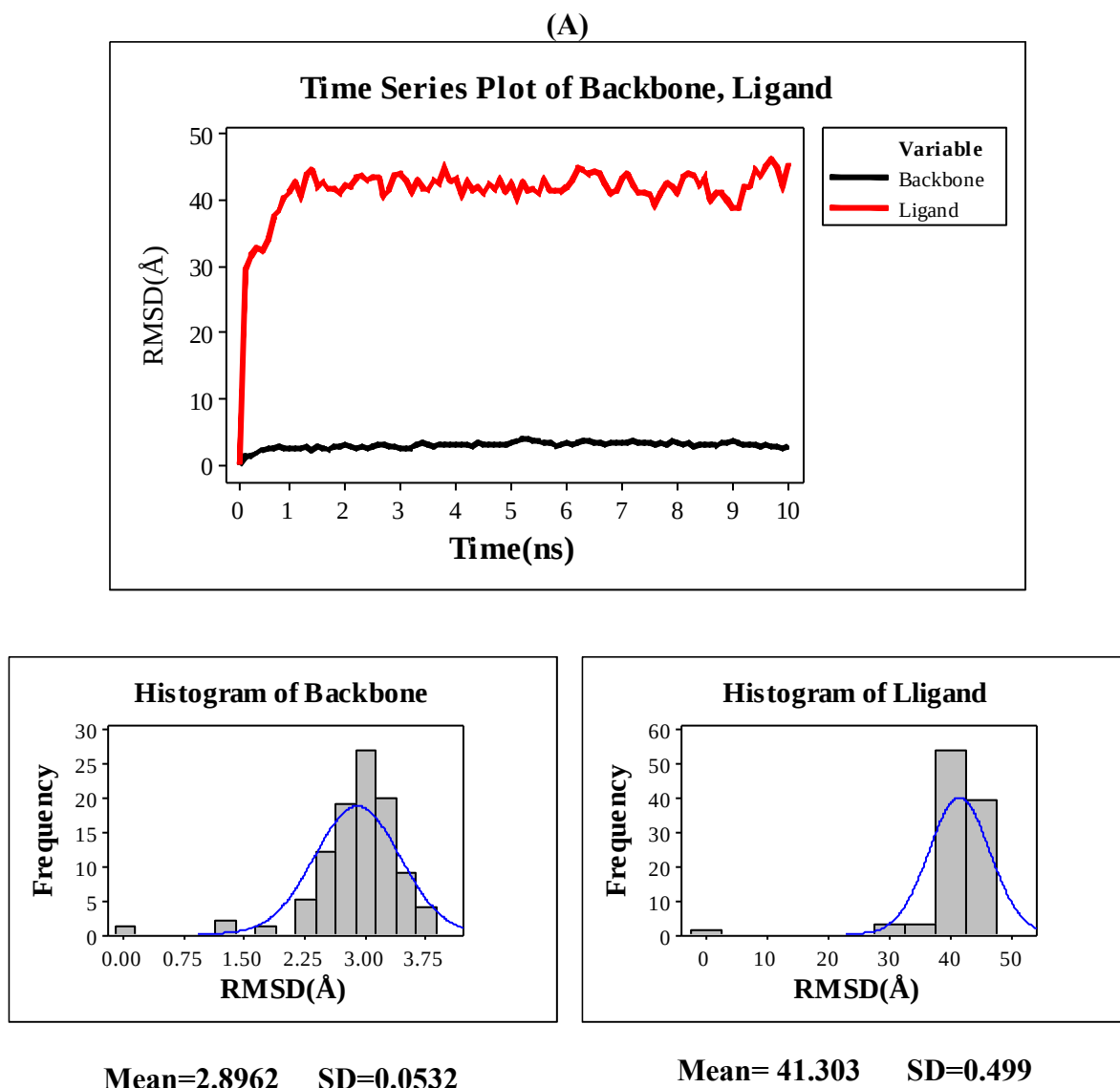
Histogram (Figure 3.5 (B)) distributions verify this contrast more clearly the narrow backbone RMSD distribution centering at around 3.2 Å, indicating very little deviation in structure, and the broad ligand RMSD histogram (Figure 3.5 (C)), centered around 42 Å, representing the ligand's flexible and dynamical nature.



**Figure 3.6** The initial conformation of Benzylisothiocyanate at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket.

In the colour code of Figure 3.6, deformations of the ligand conformation towards the red lie approximately in the first second of the simulation to the initial docking pose and in a blue one to the tenth nanosecond, revealing a quite different binding mode, being indicative that the ligand adapts and repositions inside the pocket. At the same time, the protein structure from the cyan trajectory exhibits little variation in this region, consistent with the stable RMSD of the backbone. This hybrid of quantitative and visual data suggests that the protein is structurally stable, while the ligand is quite flexible and dynamically orients itself during binding. This flexible behaviour of the ligand is essential in developing interactions in the binding pocket to maximise binding affinity and to modulate the ligand's biological activity.

#### 4.9.3 Erysolin (3080557)

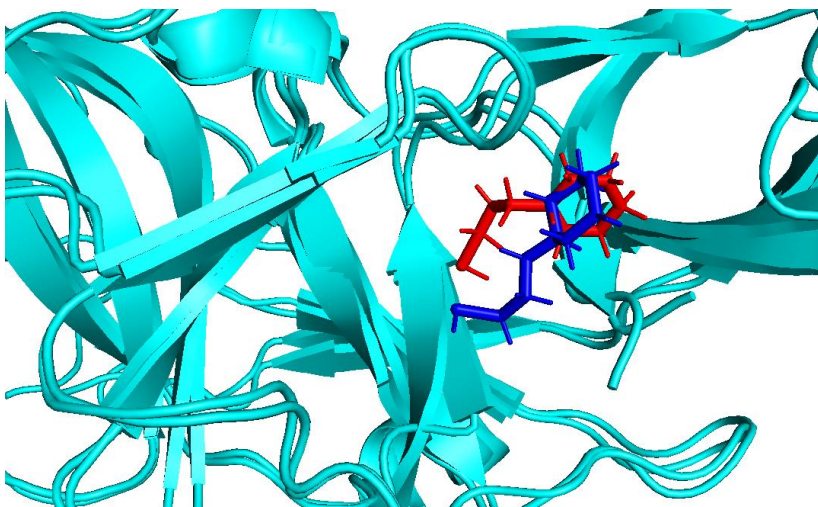


**Figure 3.7** RMSD analysis of the 10 ns molecular dynamics simulation with Erysolin (A). The time series plot of RMSD of the protein backbone (black) and ligand (red). Histograms display the distribution of RMSD values for the backbone (B) and ligand (C).

Figure 3.7 (A) illustrates the protein backbone RMSD, which stabilized at a low value of approximately 3 Å, indicating a steady overall 3D protein structure with minimal conformational rearrangements. The narrow spread of RMSD values, confirmed by a low

standard deviation (0.05), reinforces the protein's structural rigidity during the entire timeframe.

Inversely, the RMSD of the ligand increases drastically within the first nanosecond, which not only demonstrates a considerable deviation from the distinguished initial docked position but also proves its immediate loss. This increase is  $\sim 41$  Å, and it fluctuates around this high average for the remainder of the simulation, indicating large motion and conformational flexibility. The ligand has more unique conformations within the binding pocket, as indicated in the broader and higher standard deviation (0.5) of the RMSD histogram (Figure 3.7 (A)). This wide RMSD distribution illuminates the dynamic nature of the ligand and its sampling of alternate binding modes.

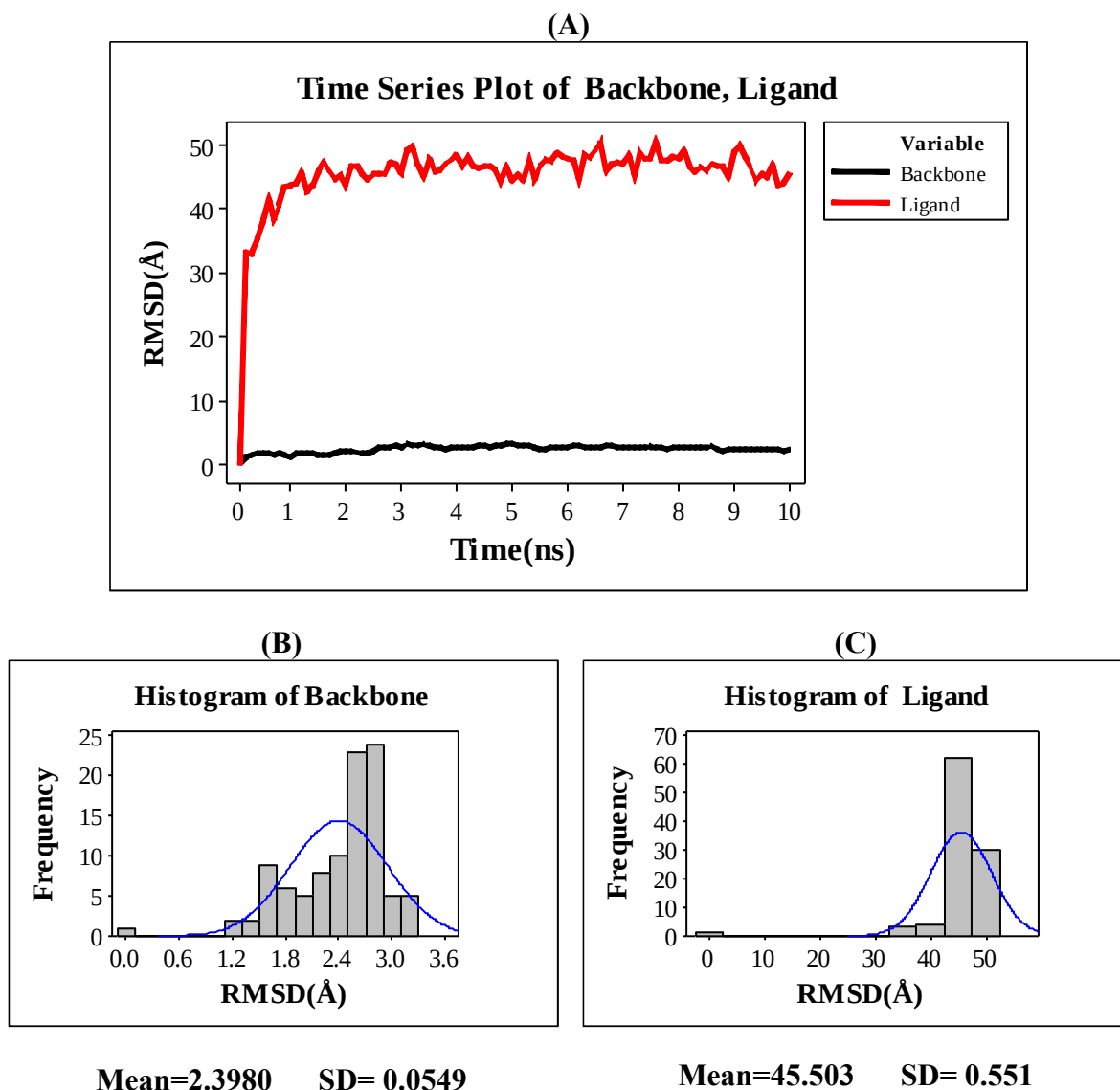


**Figure 3.8** The initial conformation of Erysolin at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket.

A quite different manner of binding is displayed in Figure 3.8, which shows the arrangement of the ligand within the pocket and its ability to adjust to fit a different pocket shape. Noting that the protein maintains structural integrity without any observable displacement, this is in agreement with the backbone RMSD results.

These quantitative and visual data indicate that, while the protein structure remains largely unchanged during the simulation, the ligand exhibits a significant degree of conformational flexibility.

#### 4.9.4 Cinnamaldehyde (637511)



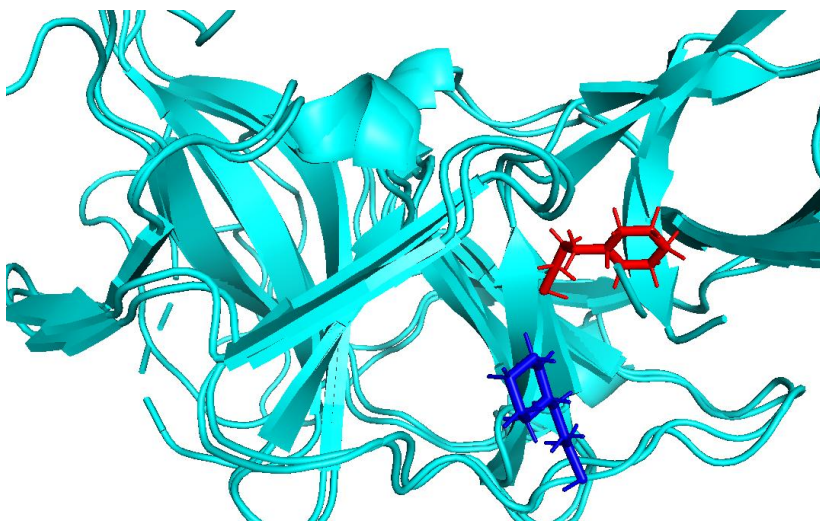
**Figure 3.9** RMSD analysis of the 10 ns molecular dynamics simulation with cinnamaldehyde (A). The time series plot of RMSD of the protein backbone (black) and ligand (red). Histograms display the distribution of RMSD values for the backbone (B) and ligand (C).

Fig. 3.9(A) shows the RMSD profile during 10ns MD simulations for cinnamaldehyde and protein. The protein backbone shows a low and relatively constant RMSD between 2.2 and 2.5 Å, with a small fluctuation ( $SD \sim 0.05$  Å), suggesting that there is no significant



conformational change in the protein shape during the simulation. In contrast, the ligand's RMSD rockets up in the first nanosecond, reaching approximately 45.5 Å, and then remains rather erratic and fluctuates greatly for the remainder of the simulation. This implies that the ligand experiences extensive conformational change, departing from the bound conformation and visiting a series of binding poses deep within the protein pocket.

The differences are illustrated more clearly in the histograms of Figures 3.9(B) and 3.9(C). The RMSD values for the backbone are well-centred around 2.4 Å, suggesting structural stability. In contrast, the ligand RMSD data are widely distributed between 0 and 45.5 Å, indicating its flexible and dynamic behavior in the simulation.

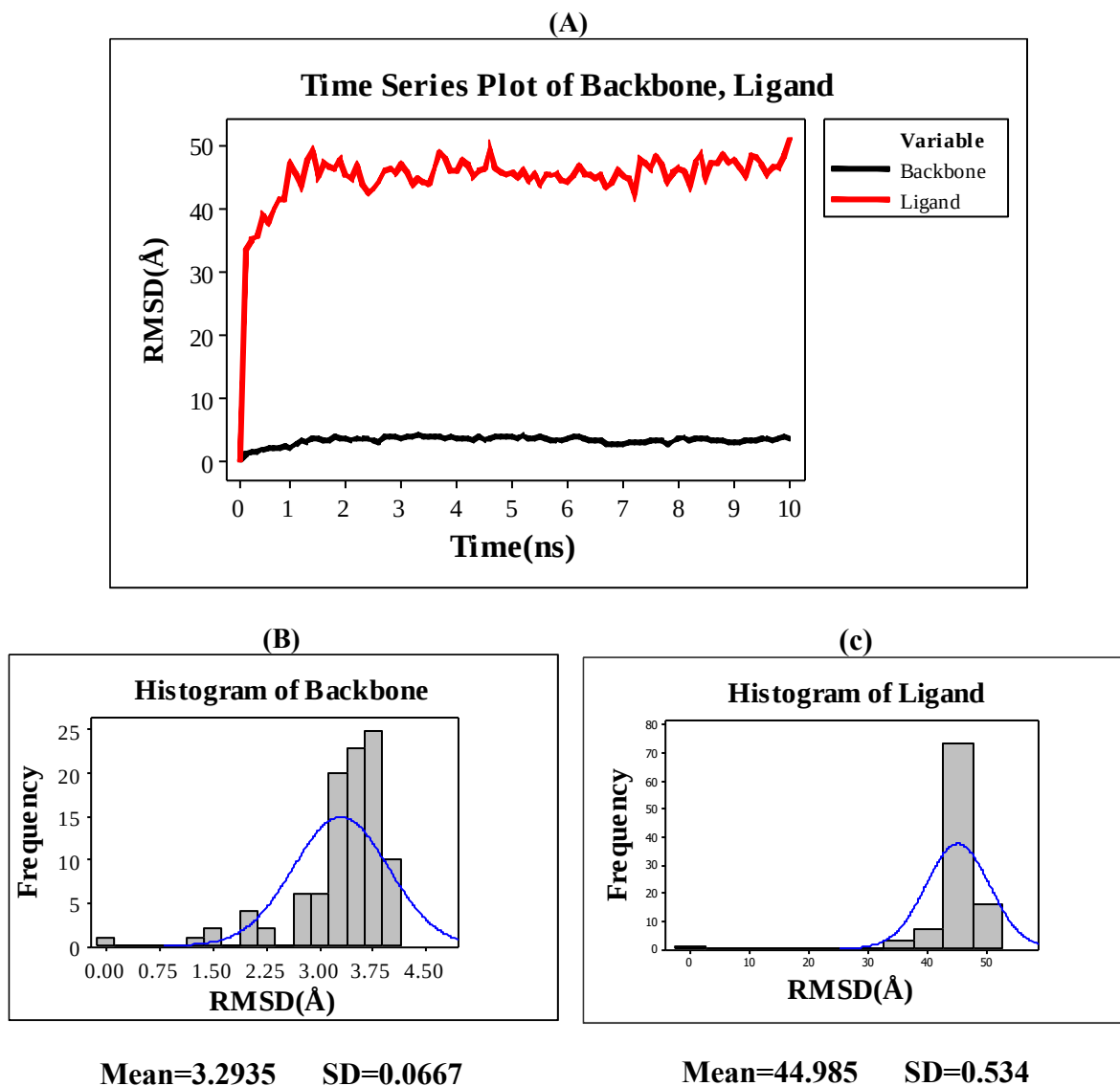


**Figure 3.10** The initial conformation of Cinnamaldehyde at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket.

These observations can be further confirmed by using a structural overlay depicted in Figure 3.10 where the red ligand represents its original binding mode at 1 ns (left) and the blue ligand represents the most changed pose at 10 ns (right), which suggests an energy-driven shift in conformation. The protein conformation (cyan), on the other hand, remains largely

preserved, consistent with the stable backbone RMSD obtained. These quantitative and visual results show that, although the shape of the protein is preserved, the ligand exhibits adaptive flexibility, thus dynamically sampling different binding modes. This movement is likely to be important in optimizing ligand-protein interactions and in enhancing binding strength and biological activity.

#### 4.9.5 Cinnamyl alcohol (5315892)



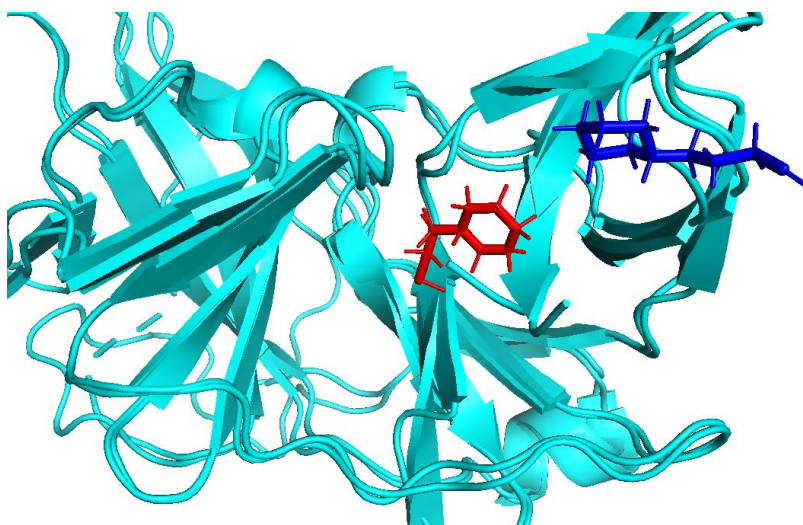
**Figure 3.11** RMSD analysis of the 10 ns molecular dynamics simulation with cinnamyl alcohol (A). The time series plot of RMSD of the protein backbone (black) and ligand (red). Histograms display the distribution of RMSD values for the backbone (B) and ligand (C).

The RMSD analysis plot is presented in Figure 3.11(A) for the 10 ns molecular dynamics simulation of cinnamyl alcohol bound to the protein. The protein backbone remains well below an average value of approximately 3.29 Å, with a fairly small standard deviation

of 0.067 Å. This indicates that the overall protein structure is stiff enough to maintain its native structure throughout the simulation.

In contrast, the ligand displays a rapid increase in RMSD during the first nanosecond, to approximately 45 Å, after which the ligand RMSD fluctuates widely about this large value, implying large conformational changes and free movement in the binding pocket. This indicates that the ligand shifts from its docked pose and samples several energetically accessible conformations within the protein active site.

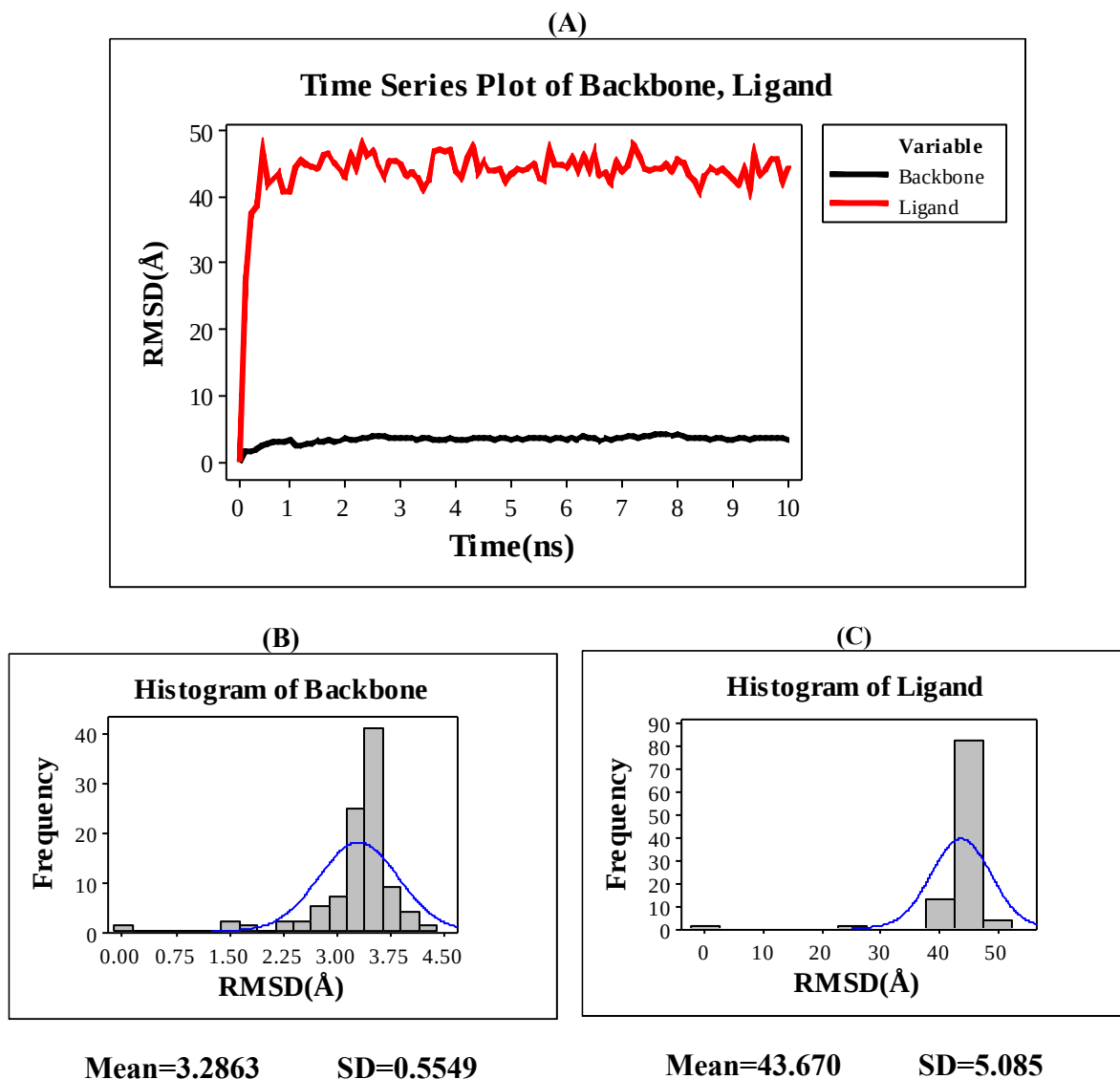
The histograms in Fig. 3.11(B) and (C) also illustrate this difference in flexibility. Both structures have a small protein backbone RMSD, which is tightly distributed around the average of 3.29 Å, indicating a high degree of structural overlap. Ligand RMSD values are also broadly distributed to a RMSD mean of 44.98 Å, indicating that the ligand is quite dynamic and flexible throughout the simulation.



**Figure 3.12** The initial conformation of Cinnamyl alcohol at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket.

These quantitative results are illustrated in Figure 3.12, showing two ligand conformations of interest: the red ligand conformation represents the initial binding conformation (1<sup>st</sup> ns), and the blue ligand conformation corresponds to the binding mode (10<sup>th</sup> ns) in which the ligand position within the protein binding pocket has changed, suggesting a large motion. The structure of the protein, represented in cyan, is largely invariant and correlates with the low root mean square deviation (RMSD) of the backbone. These data show that the protein shows a stable 3D structure, while the ligand has a high degree of flexibility, adjusting dynamically its binding mode to maximize interactions with the protein. Mobile ligands could be crucial in achieving strong and specific molecular recognition, which may lead to enhanced biological activity.

#### 4.9.6 Castanospermine (54445)



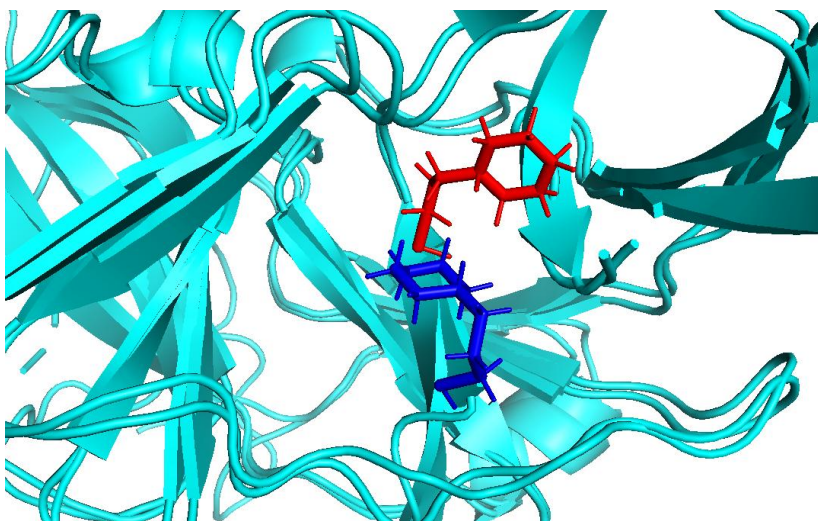
**Figure 3.13** RMSD analysis of the 10 ns molecular dynamics simulation with castanospermine (A). The time series plot of RMSD of the protein backbone (black) and ligand (red). Histograms display the distribution of RMSD values for the backbone (B) and ligand (C).

The RMSD time series of the protein backbone and ligand during a 10 ns molecular dynamics simulation with castanospermine are shown in Figure 3.13 (A). The protein backbone RMSD remains low and relatively constant, with an average value of around 3.29 Å

(SD  $\sim 0.55$  Å), indicating that the global shape of the protein is very stable during the simulation.

In contrast, the ligand's RMSD shoots up very rapidly in the 1st nanosecond to a value of approximately 44 Å, after which it fluctuates about this high value with a standard deviation of  $\sim 5.1$  Å for the remaining duration of the simulation. That implies the ligand is free to move around a considerable amount in the binding pocket, changing both its shape and position, rather than being locked in place.

The (B) and (C) histograms in Figure 3.13 support this explanation. The distribution of the backbone RMSD is narrow and is centered around 3.3 Å, indicating small changes in conformation. On the other hand, the ligand RMSD histogram is broader, with a peak at 43.7 Å, indicating that it remains widespread and floppy throughout the entire simulation time.



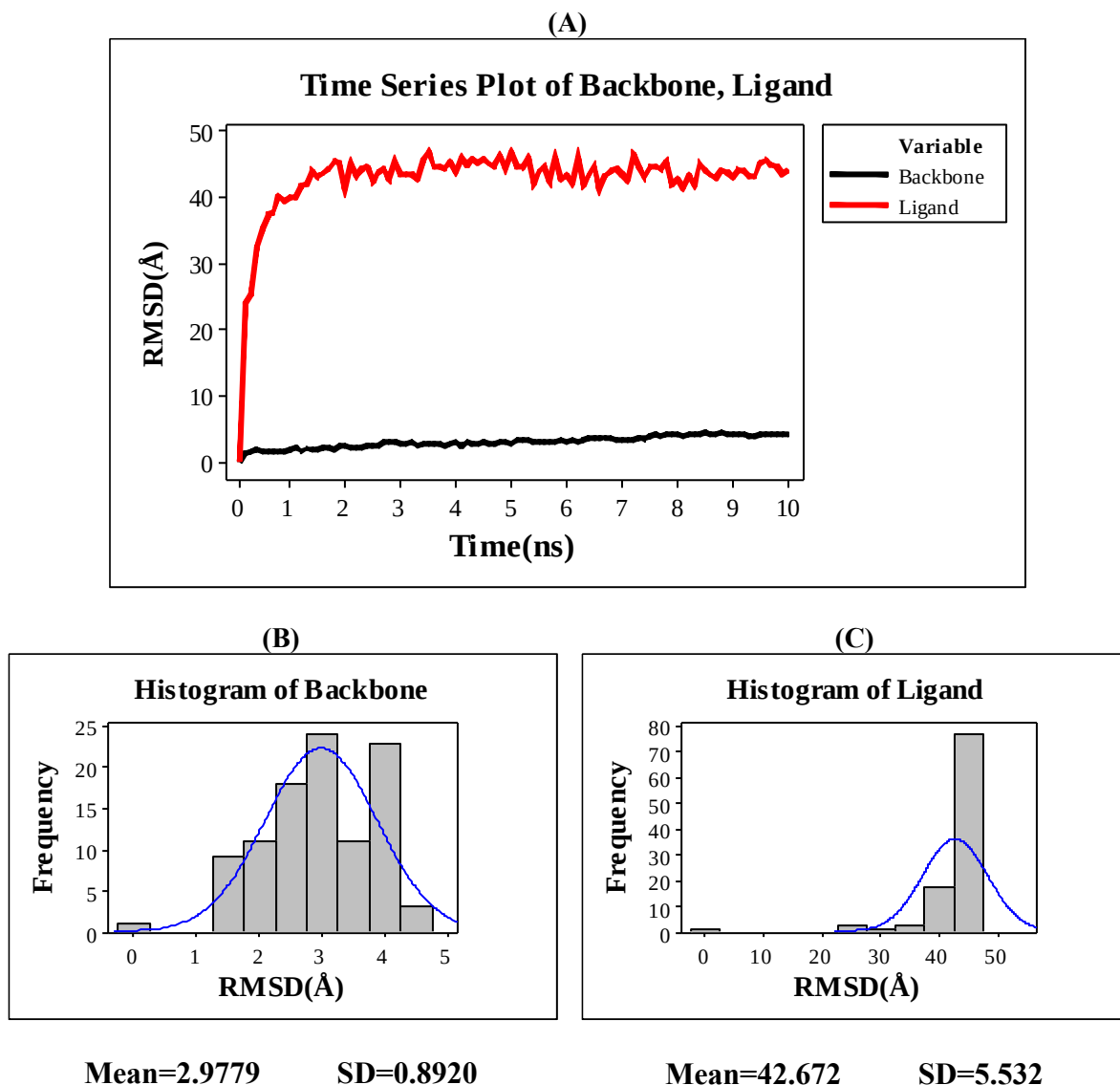
**Figure 3.14** The initial conformation of Castanospermine at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket.

In Figure 3.14 the ligand conformation in red shows the same original pose as the ligand's starting position at the beginning of the simulation. The blue ligand conformation at

10 ns represents a strikingly different binding mode, demonstrating how the ligand can accommodate itself and move within the pocket. The protein, drawn in cyan, preserves its structure, which is supported by low backbone RMSD values. The protein rigidity and stability as compared to the high mobility and flexibility of castanospermine, which undergoes major conformational changes within the binding site. This adaptability may, in turn, enable ligands to better adjust their interactions, including stronger binding and control of their biological activity.



#### 4.9.7 Cinnamic Acid (444539)



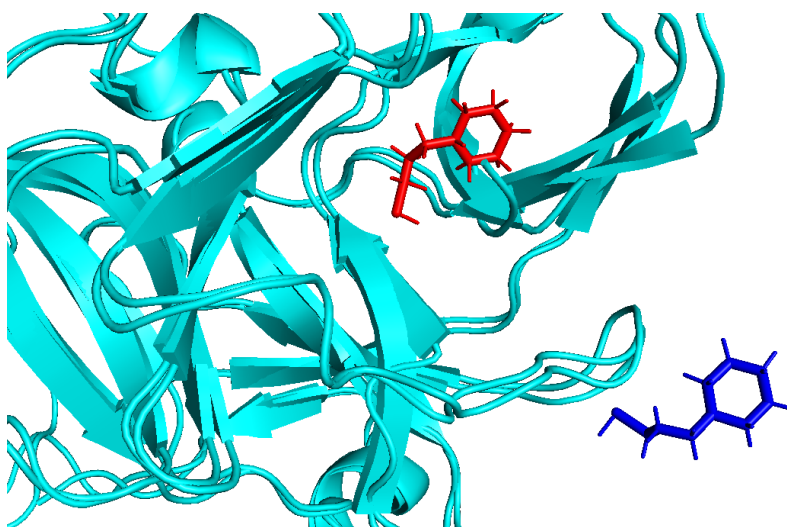
**Figure 3.15** RMSD analysis of the 10 ns molecular dynamics simulation with cinnamic acid(A). The time series plot of RMSD of the protein backbone (black) and ligand (red). Histograms display the distribution of RMSD values for the backbone (B) and ligand (C).

The RMSD values of the scales and the backbone atoms of the ligand castanospermine during the MD simulation are presented in Figure 3.15 (A). The protein backbone RMSD is low and

stable, with an average value of 3.29 Å (SD 0.55), indicating that the protein retains its global structure and does not undergo large conformational changes during the simulation.

In contrast, the ligand increases its RMSD quickly within the first nanosecond, reaching approximately 44 Å, and subsequently moves around this high RMSD value with a standard deviation of around 5.1 Å, indicating that the ligand is very mobile in the binding pocket and changes its conformation and location over time rather than remaining static.

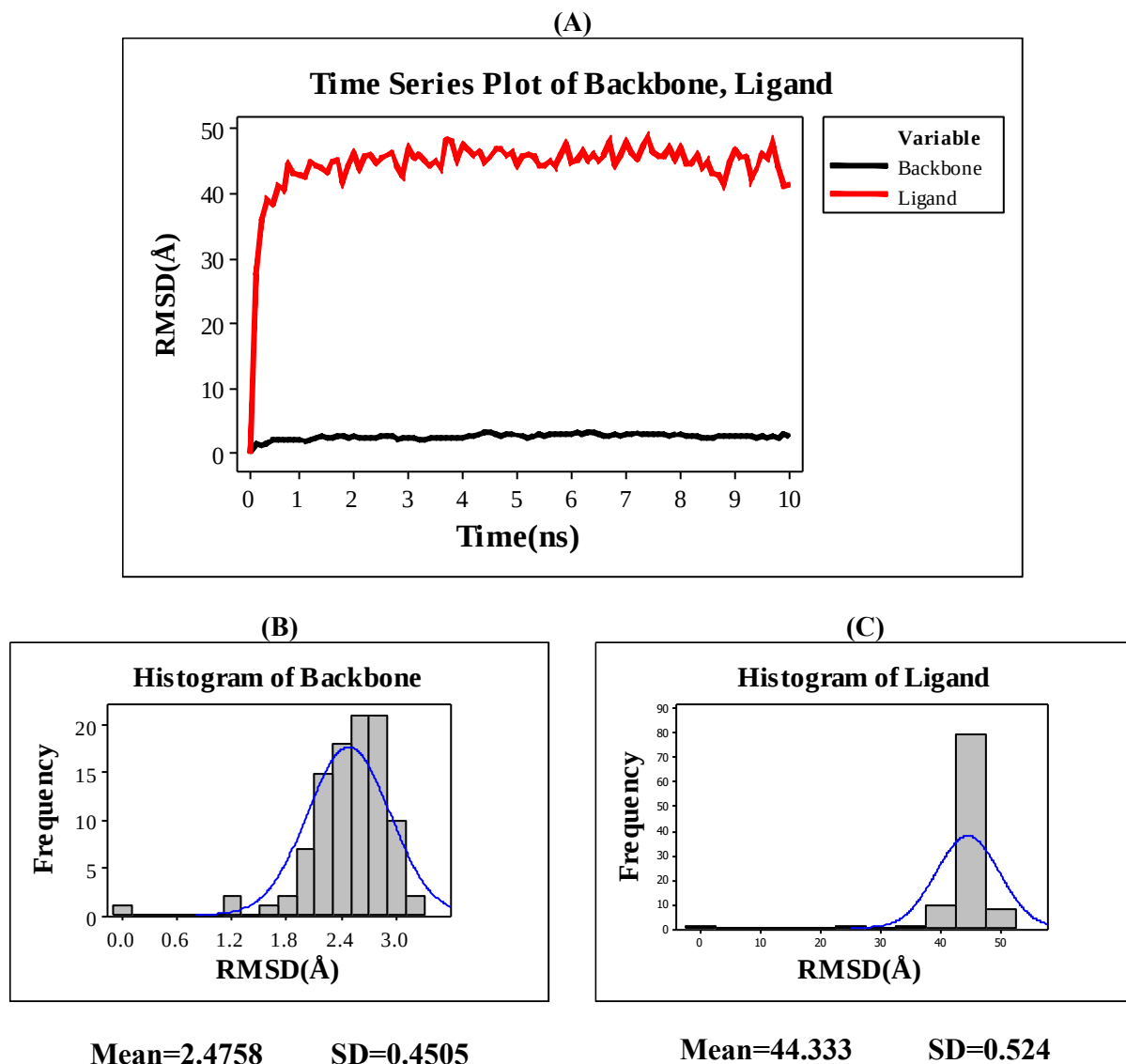
This observation is corroborated by the histograms represented in Figures 3.15 (B) and (C). The distribution of backbone RMSDs is narrow and strongly peaked around 3.3 Å, showing that the shape of the protein is stable. The ligand RMSD distribution, on the other hand, is wide, centered at 43.7 Å, which indicates its flexibility, as the ligand can sample multiple conformations during the simulation.



**Figure 3.16** The initial conformation of Cinnamic Acid at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket.

It is clearly observed in Fig. 3.16 that the ligand adopts a different mode of binding within the pocket. The protein, on the other hand, in cyan, keeps a very clear shape, which goes in line with the non-fluctuating RMSD. while the protein backbone remains rigid and stable, castanospermine exhibits significant movement and conformational changes within the binding site. This flexibility likely allows the ligand to adjust and optimize its interactions with the protein, which could enhance binding strength and impact its biological function.

#### 4.9.8 Sulforaphane (5350)

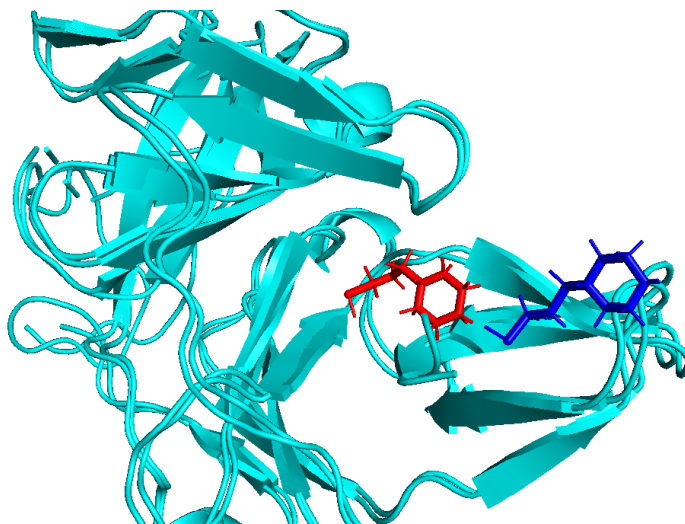


**Figure 3.17** RMSD analysis of the 10 ns molecular dynamics simulation with sulforaphane (A). The time series plot of RMSD of the protein backbone (black) and ligand (red). Histograms display the distribution of RMSD values for the backbone (B) and ligand (C).

The RMSD time series of a molecular dynamics simulation of sulforaphane binding to a protein is plotted in Figure 3.17(A). The protein backbone RMSD remains small and constant (approximately  $2.48 \pm 0.4547$ ), which means that during the simulation, the structure of the protein does not change much, which means it is quite rigid and stable. In contrast, the ligand

RMSD (red line of Figure 3.17 (A)) quickly jumps to the value around 44.33 Å within the first nanosecond and oscillates around the high value with a low standard deviation of 0.52 Å during the simulation. This demonstrates that the ligand is quite mobile, it changes conformation and does not sit still in one single pose.

Histograms of the RMSD for Figures 3.17 (B) and (C) are presented. The histogram for the protein backbone also shows a small, single peak at approximately 2.48 Å, supporting the conclusion that the protein undergoes very few structural changes during the simulation. In contrast, the ligand RMSD histogram is distributed over a wide range centered at 44.33 Å, indicating that the ligands are highly flexible and mobile within the binding site.



**Figure 3.18** The initial conformation of Sulforaphane at 1<sup>st</sup> nanosecond (red) and its repositioned conformation at 10<sup>th</sup> nanoseconds (dark blue) within the protein binding pocket.

These conclusions are supported by the visual 3D structural overlay in Figure 3.18. The red ligand is superimposed where it was docked at 1 ns and is distinctly different, highlighting that the ligand had shifted its binding pose to a more stabilizing position. The protein backbone (cyan), however, remains almost similar, consistent with the steady RMSD plot. The backbone

stability of the protein with ligand flexible exploratory behavior indicates that the ligand is actively searching for the optimal binding fit, which may enhance the binding affinity and biological function of Sulforaphane with the target protein.

In this thesis, 10 ligands MM/GBSA, MM/PBSA and RMSDs were calculated, but the last two docking values (least binding affinity in docking) were neglected due to the MM/GBSA and the MM/PBSA value gave good values, but in the MD (simulation run for 10 ns), the ligand moved away from the binding pocket. The data were excluded here.

#### **4.10 H-Bond Analysis**

Hydrogen bonding plays a critical role in molecular recognition, structural stability and biological specificity. In structure-based drug design, binding affinity and selectivity are highly affected by hydrogen bonds between the ligand and its target protein. Understanding about the hydrogen bonds formed in the ligand-receptor can offer a valuable idea about the strength and stability of the interaction.

Hydrogen bond analysis helps to find how the ligand fits into the active site in the protein and find the natural inhibitors. These bonds appear polar groups such as hydroxyl, amine or carbonyl moieties and are classified by directionality and distance constraints to make them for accurate binding pose predictions.

In this study, hydrogen bond analysis was performed on the best 8 ligands with NS2B-NS3 protease. The number, position and nature of the hydrogen were examined to find the inhibitory ligands to support the selection of candidates with favorable binding characteristics.

**Table 3.3** Table of H-bond analysis of the Molecular Dynamics simulation of protein and ligand complexes.

| Molecule Name               | Major H-bond Residue | Occupancy |
|-----------------------------|----------------------|-----------|
| Salianic acid (11600642)    | ALA195               | 47.52%    |
|                             | ASN181               | 14.85%    |
| Benzylisothiocyanate (2346) | ASN196               | 14.85%    |
|                             | ASN181               | 13.86%    |
|                             | LYS103               | 5.94%     |
| Erysolin (3080557)          | LYS133               | 82.18%    |
|                             | SER160               | 75.25%    |
|                             | GLN64                | 70.30%    |
|                             | THR163               | 65.35%    |
|                             | SER156               | 64.36%    |
| Cinnamaldehyde (637511)     | VAL81                | 72.28%    |
|                             | VAL69                | 71.29%    |
|                             | VAL191               | 69.31%    |
|                             | VAL183               | 67.33%    |
|                             | VAL176               | 66.34%    |
| Cinnamyl alcohol (5315892)  | LYS133               | 82.18%    |
|                             | SER160               | 73.27%    |
|                             | GLN64                | 70.30%    |
|                             | THR163               | 68.32%    |
|                             | SER156               | 67.33%    |
| Castanospermine (54445)     | LYS133               | 80.20%    |
|                             | SER160               | 74.26%    |
|                             | GLN64                | 70.30%    |
|                             | THR163               | 68.32%    |
|                             | SER156               | 68.32%    |
| Cinnamic Acid (444539)      | LYS133               | 74.26%    |
|                             | SER160               | 70.30%    |
|                             | GLN64                | 70.30%    |
|                             | THR163               | 66.34%    |
|                             | SER156               | 66.34%    |
| Sulforaphane (5350)         | LYS133               | 74.26%    |
|                             | SER160               | 68.32%    |
|                             | GLN64                | 67.33%    |
|                             | THR163               | 66.34%    |
|                             | SER156               | 64.36%    |

Hydrogen bonds are crucial in the stability and specificity of protein-ligand complexes during molecular dynamics (MD) simulations. These interactions can be quantitatively monitored based on H-bond occupancy (the percentage of frames during a simulation in which a given H-bond is formed between a ligand and a protein residue). Generally, an occupancy greater than 5% is considered significant, indicating a stable, potentially functionally relevant interaction. The table shown is limited to hydrogen bonds that exceed these constraints and consequently iterates on meaningful contacts that mediate the stability of ligand binding.

From the ligands analyzed, several common residues appeared as major contributors to the high-occupancy hydrogen-bond network. Specifically, LYS133 (Sidechain) had stable occupancy for both Erysolin (82.18%), Cinnamyl alcohol (82.18%), Castanospermine (80.20%), Cinnamic Acid (74.26%) and Sulforaphane (74.26%). This identifies LYS133 as a central docking residue in the binding pocket. Also, SER160 (Main), GLN64 (Main), THR163 (Side), and SER156 (Side) showed high-occupancy interactions with several ligands, which suggests their significant contributions to the stabilization of the ligand-protein complex.

Notably, Erysolin exhibited the greatest cumulative H-bond occupancies, indicating particularly stable binding-site interactions. Both cinnamyl alcohol and Castanospermine formed hydrogen bonds well almost all the time, suggesting they are also stable binders. By comparison, Salaniac acid and Benzylisothiocyanate also possessed fewer and less occupancy of the H-bonds, indicating their interaction towards the protein target was weaker.

In summary, the detection of defined high-occupancy interactions across multiple ligands indicates the conservation of a hydrogen bonding network for optimal binding. Ligands that mimic this network, in particular, with groups such as LYS133, SER160 and THR163, are



likely to display stronger binding and stability, and hence, would be good leads for further drug design against this protein.

#### **4.11 Toxicity Predictions**

The toxicity profiles of 8 selected ligand molecules were analyzed using ProTox 3.0, an advanced in silico toxicity prediction platform available at <https://tox.charite.de/protox3/>. ProTox 3.0 combines molecular similarity searches with machine learning models to estimate a broad range of 61 toxicity endpoints. These categories are acute toxicity, organ toxicity, clinical toxicity, molecular initiating events, adverse outcome pathways, and off-target toxicity. The 2D chemical structure of each ligand was downloaded into the tool, which provided predictions, including the LD<sub>50</sub> value in terms of mg/kg, the associated toxicity class according to the Globally Harmonized System (GHS), and potential activity on various toxicological endpoints. The platform also includes confidence scores, toxicity radar charts, and network maps to provide the graphical outputs of the predicted toxicological profile. This integrated approach facilitates the early detection of drugs with undesirable safety attributes and simplifies the selection of optimal candidates with the best therapeutic potential.

**Table 3.4** Description of toxicity classes.

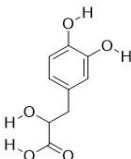
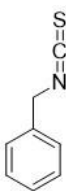
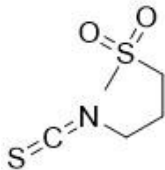
| Drug category | Description                                             |
|---------------|---------------------------------------------------------|
| Class 1       | Fatal if swallowed ( $LD50 \leq 5$ )                    |
| Class 2       | May be fatal if swallowed ( $5 < LD50 \leq 50$ )        |
| Class 3       | Toxic if swallowed ( $50 < LD50 \leq 300$ )             |
| Class 4       | Harmful if swallowed ( $300 < LD50 \leq 2000$ )         |
| Class 5       | May be harmful if swallowed ( $2000 < LD50 \leq 5000$ ) |
| Class 6       | Non-toxic ( $LD50 > 5000$ )                             |

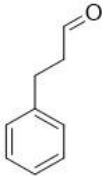
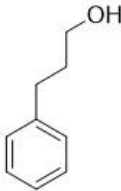
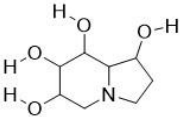
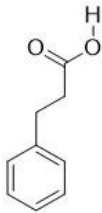
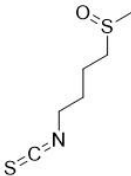
The LD50 value is a quantitative measure of a drug's toxicity, which can be used to classify drugs into different categories based on their lethal dose 50% (LD50) value. The LD50 is defined as the dose that kills 50% of the test animals and is typically measured in milligrams per kilogram of body weight. There are six classes of drugs categorized based on their LD50 values, which indicate the lethal dose required to cause harm.

Table 3.4 illustrates the toxicity classes and their description. According to that, Class 1 drugs are drugs with an LD50 value of 5 or less that are considered fatal, and ingesting even a small amount can be lethal. Drugs with LD50 values between 5 and 50 pose a significant risk and can cause severe harm if ingested. It is known as a class 2 drug. Class 3 Drugs are drugs with LD50 values between 50 and 300, and are classified as toxic if ingested, and they have the potential to be harmful to one's health. Drugs in class 4 have LD50 values between 300 and 2000, and are dangerous if ingested, and they can lead to adverse effects and potential health risks. Besides that, Class 5 Drugs are drugs with LD50 values between 2000 and 5000 that may be harmful if swallowed, and ingesting these drugs can lead to adverse effects and

potential health risks. Finally, Class 6 Drugs with LD50 values higher than 5000 are considered non-toxic, as they are unlikely to cause harm if ingested. Table 3.5 shows the toxicity of the selected best ligands.

**Table 3.5** List of molecules with their Molecular IDs, chemical structures, and corresponding predicted toxicity classes.

| No | Molecular ID | Molecule                                                                             | Predicted toxicity class |
|----|--------------|--------------------------------------------------------------------------------------|--------------------------|
| 1  | 11600642     |    | 5                        |
| 2  | 2346         |   | 4                        |
| 3  | 3080557      |  | 4                        |

|   |         |                                                                                      |   |
|---|---------|--------------------------------------------------------------------------------------|---|
| 4 | 637511  |     | 4 |
| 5 | 5315892 |     | 4 |
| 6 | 5445    |    | 4 |
| 7 | 444539  |   | 5 |
| 8 | 5350    |  | 4 |

Toxicity prediction of the 8 selected ligand molecules was performed using proTox 3.0, a web-based tool for the prediction of toxic effects of chemicals from their structural formula. The tool classifies chemicals into toxicity classes, using the GHS, from Class 1 (most toxic) to

Class 6 (non-toxic). According to ProTox 3.0 predictions, most of the compounds fell into the Toxicity Class 4, moderate poisons category. In particular, inhibitors with molecular IDs 2346, 3080557, 637511, 5315892, 5445, 5350, and 6199 all belong to this class of inhibitors. Meanwhile, chemicals 11600642, 444539, and 323 were classified in Toxicity Class 5, indicating their low toxicity. These results suggest that none of the ligands is highly toxic (classes 1-3) and can be viewed as relatively safe for further use in drug discovery research. The classification offers a preliminary insight into the safety profile of each compound, and its application is useful for filtering out less toxic ligands for further biological assay and optimization.

#### 4.12 Conclusion

The post-docking binding free energy calculation using MM/GBSA and MM/PBSA indicated some suggestive features regarding the binding affinities of the top 10 ligand candidates against dengue virus NS2B-NS3 protease. The calculated ligand binding energies were highly variable, with ligand 5445 exhibiting the weakest interaction ( $-0.69 \pm 5.25$  kcal/mol) and ligand 3080557 the strongest ( $-21.43 \pm 1.70$  kcal/mol), demonstrating a broad range of predicted binding strengths. Ligands such as 3080557, 444539, and 5315892, although exhibiting moderate docking scores, were consistently found to be better binders in MM/GBSA and MM/PBSA analysis, indicating that these molecules might have better stability and a preferred interaction profile than predicted from docking results. This was also substantiated by hydrogen bond occupancy analysis, which showed that important residues, such as LYS133, SER160, GLN64, THR163, and SER156, displayed high occupancy values with most ligands, suggesting stable and consistent interactions necessary for effective inhibition. Toxicity prediction showed that for the majority of the top ligands, they are classified as risk class 4, indicating moderate toxicity; only ligands 11600642 and 444539 are classified as risk class 5, indicating a lower toxicity risk. All in all, the ligands 3080557, 444539, and 5315892 offer promising leads in terms of good binding affinity, stable molecular interactions, and manageable toxicity. They demand in vitro and in vivo testing to confirm their value for the dengue virus protease.

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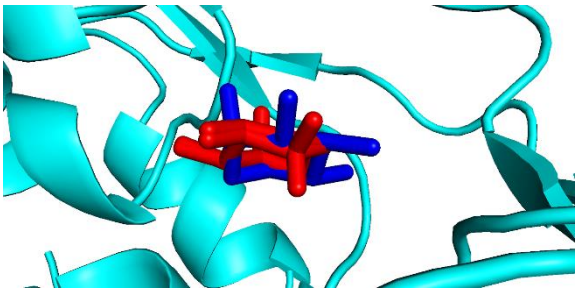
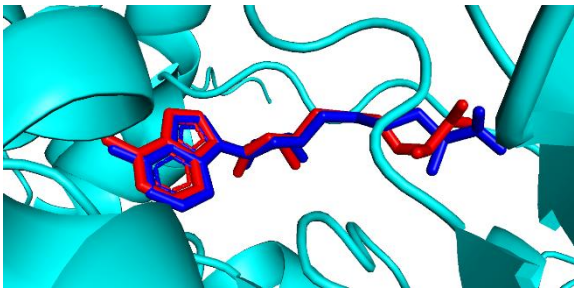
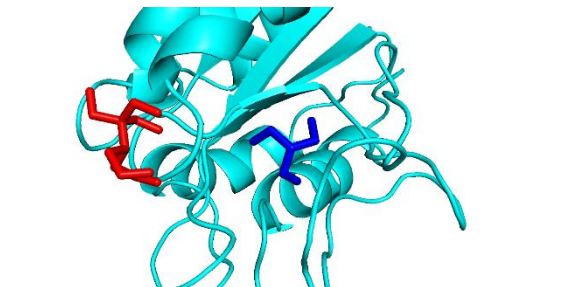
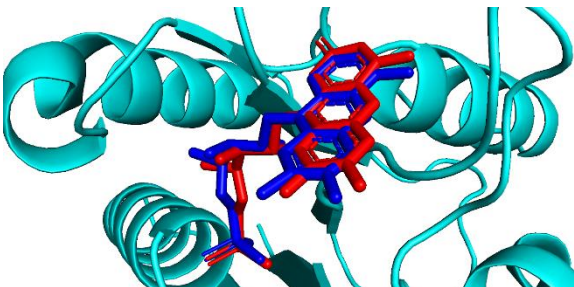
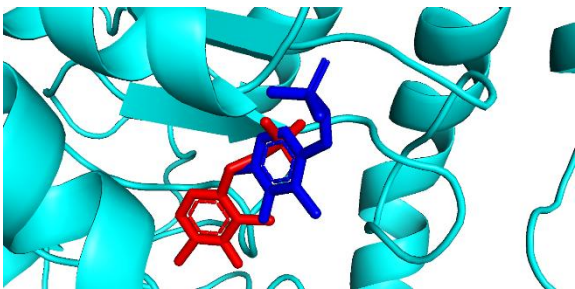
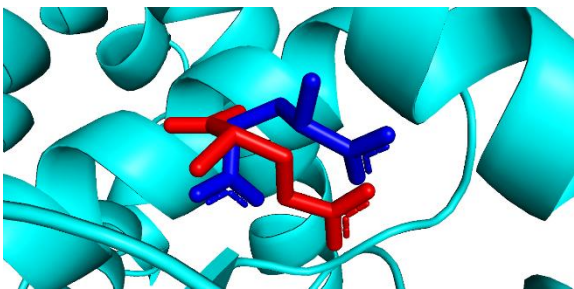
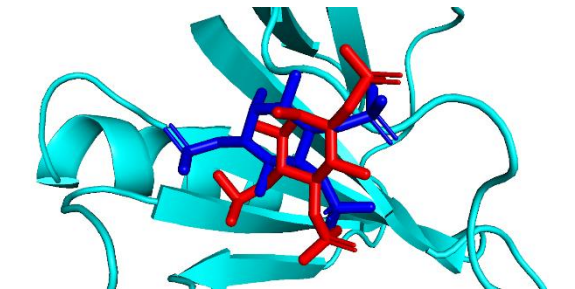
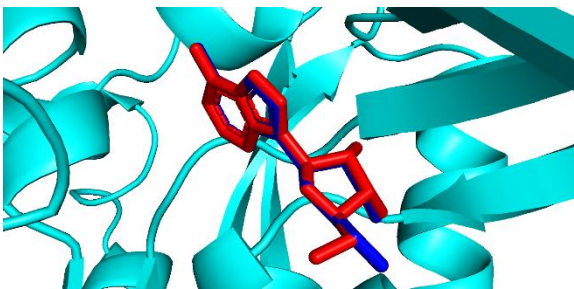
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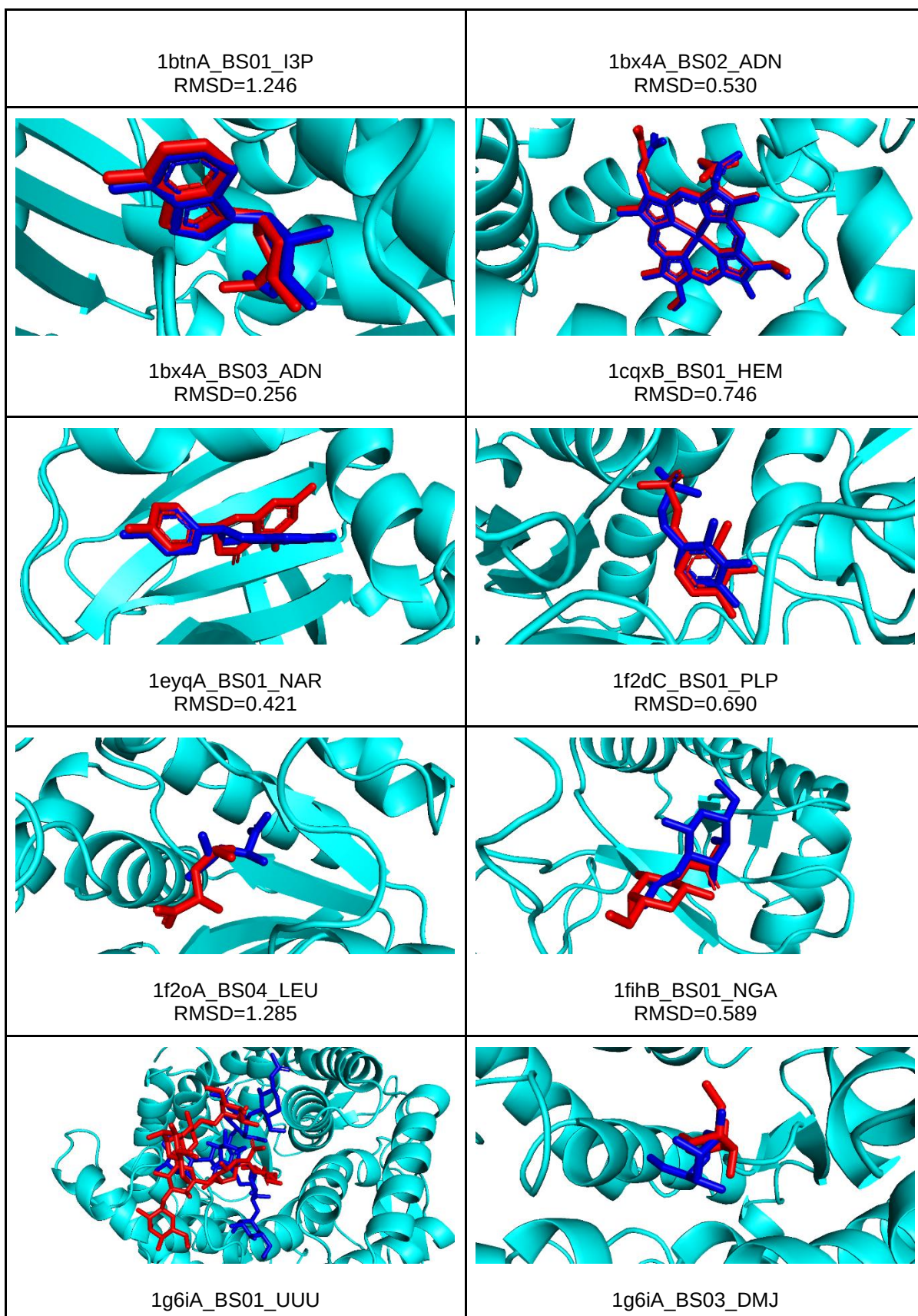
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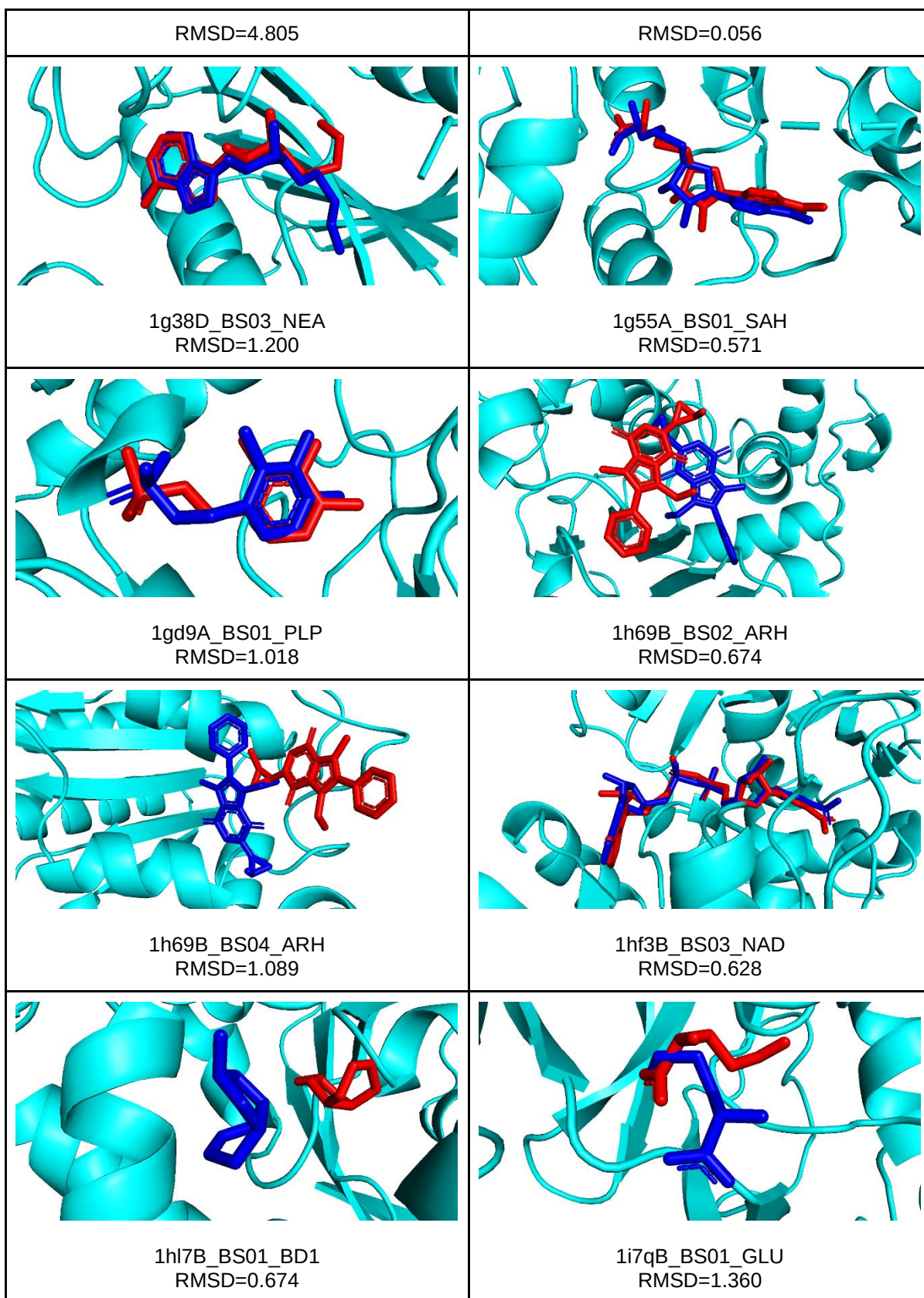
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## APPENDIX

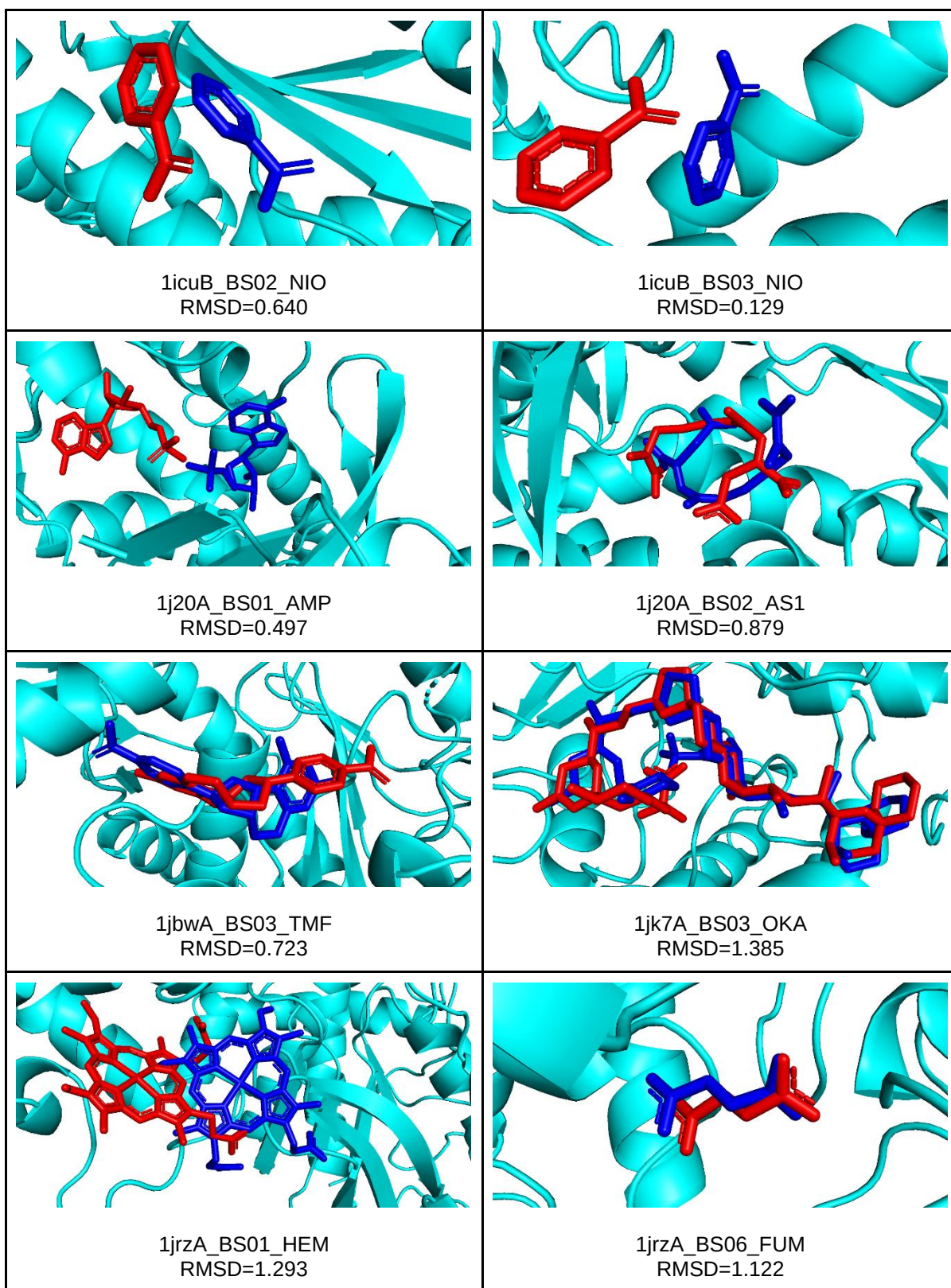
| Table of visual representation of benchmark test using Galaxy BRIDGE Tool                                                |                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|  <p>1abeA_BS01_UUU<br/>RMSD=0.044</p>   |  <p>1af7A_BS01_SAH<br/>RMSD=0.633</p>   |
|  <p>1ag9B_BS04_BTBT<br/>RMSD=0.771</p> |  <p>1akuA_BS01_FMN<br/>RMSD=0.333</p>  |
|  <p>1b9hA_BS01_PLP<br/>RMSD=1.659</p> |  <p>1bgvA_BS01_GLU<br/>RMSD=1.217</p> |
|                                       |                                       |

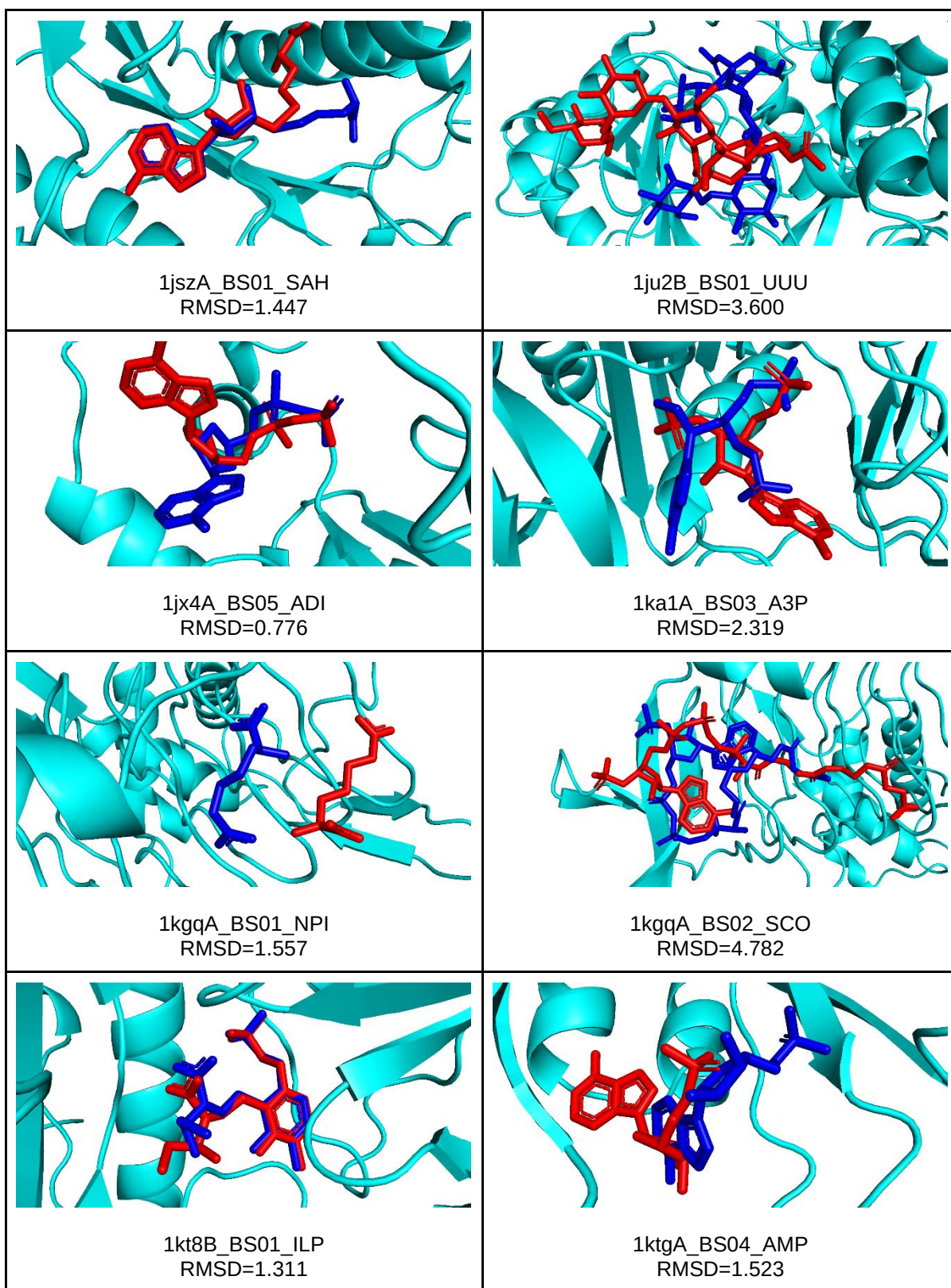












## DATA AVAILABILITY

[https://drive.google.com/drive/folders/1QLWrFqUrHlCEafPEyJLBTf7g\\_RtEGQYh?usp=sharing](https://drive.google.com/drive/folders/1QLWrFqUrHlCEafPEyJLBTf7g_RtEGQYh?usp=sharing)